Project 3: Ocean Map

Documentation set, October 9, 2026

This PDF is built from the files in the Ocean Map folder. Links to other folders appear as plain text.

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Part 1: Overview

`Ocean Map/README.md`

Project 3: Ocean map. README

What it is. Enter an ocean or coastal point and see how much nearby seafloor is unmapped, the fishing activity nearby, and the nearest buoy's latest reading with the time it was taken. In Phase B the project adds a live vessel stream, a cleaned vessel identity table with a human review queue, a classifier for vessel matches, and buoy fault flags. Phase C scales the work to a full year of AIS data.

Status. Design stage. The documents in this folder are drafts awaiting John's approval. No code exists yet, and there is no live link. The Phase A target is a public lookup that answers with the session box off.

Live link. None yet. It is added when Phase A is deployed.

The documents

Document	What it holds
requirements.md	Who the project is for, features by phase, functional and quality requirements, acceptance checks, and the proposed step list (section 8)
design-doc.md	Architecture, components, data model, API, deployment, cross-cutting concerns, testing, alternatives, risks, open questions (section 16) and glossary
adrs/	One record per significant decision, with status, context, alternatives and consequences
contracts/	A data contract for every source and every table in ODCS format
openapi.yaml	The API specification for the lookup and the owner-only review routes
operations.md	Schedules, freshness, monitoring, cost limits and runbooks
security-and-privacy.md	Threats, keys, personal-data inventory and the licence and credit register
model-card.md	The AI parts: intended use, evaluation, known limits and safety
changelog.md	What changed in each revision
decisions.md	John's answers and this project's decision log

The shared rules for the whole project (repository and CI, cloud base, the nightly session box, the serving template, status and incidents, and the operating rules) are in the platform set: Platform/README.md. These documents link to it rather than repeat it.

Before you set it up

You need these accounts and keys. Each key is stored in AWS Parameter Store and read by role. None is written into code or into GitHub.

- An AWS account and the platform's cloud base (platform step 0.2).
- A Global Fishing Watch API key. The API is free for non-commercial use and needs an account (see the credit and licence rules in security-and-privacy.md).
- An AISStream API key (Phase B). The service is free. Its terms are not yet verified (open question OQ-3).
- A model API key, with the monthly spend limit set in the provider account before the first call (platform decision on the model API).
- Databricks Free Edition and a Snowflake trial, both used in Phase B. Classifier results are exported by hand and committed to GitHub.

Setup commands. None yet. They are written when the code exists, starting with platform step 0.1 (repository and CI).

Tests. Described in design-doc.md section 13. They are not written yet.

Credit and licences

- Global Fishing Watch data: non-commercial use only (CC BY-NC 4.0). The page says "Powered by Global Fishing Watch", linked to globalfishingwatch.org. Verified on the provider's license page (see security-and-privacy.md).
- AISStream: the page shows only maps and counts with credit, and no vessel names or MMSI (ADR 0008). The provider's terms are not verified yet.
- GEBCO: public domain grid; credit is required, with the exact wording from the documentation that comes with the grid. That wording is not yet verified (OQ-24).
- NOAA NDBC and MarineCadastre: credit wording and licence status are not yet verified (OQ-24). See security-and-privacy.md.

Open questions

The open questions and their answers are in design-doc.md section 16. Answered questions stay in that table with the answer and the ADR number.

Part 2: Requirements

`Ocean Map/requirements.md`

Project 3: Ocean map. Requirements

Field	Value
Status	Draft for John's review and approval (not yet approved)
Owner	John
Reviewers	John
Last updated	Oct 9, 2026
Sources	Build Plans for the Five Projects (Oct 8, 2026), Build Plan Decisions (Oct 9, 2026), Documentation Plan (Oct 9, 2026)
Related	design-doc.md, adrs/, contracts/, openapi.yaml, platform documents (written first; draft awaiting John's approval)

Conventions. Every value here is a recorded decision (cited by ADR number, see adrs/) or a confirmed fact from the three source documents. Anything not determined is listed in section 16 of the design doc as an open question, referenced here as OQ-n. Requirements that the build plan does not state but that follow from the platform's engineering standards are marked "(platform rule n)". Requirements that are proposals by this document and need John's approval are marked "(proposed)" and also listed in the section "Proposed additions" at the end.

Terms and abbreviations used in this document

The full glossary is in design-doc.md section 17.

Term	Meaning
The plan, build plan	The document "Build Plans for the Five Projects"
Done when	The build plan's completion check for a step
Session box	The temporary AWS server that runs the pipelines; it is on during work sessions and during the nightly run, and off otherwise
Nightly run	The scheduled run of every pipeline, started at 10:00 UTC
Lookup	The public answer for one ocean or coastal point
Platform rule n	Engineering standard n in the build plan's "Engineering standards" table. The rules cited here: 1 keep raw files immutable with source URL, fetch time and hash (a manifest for public archives that never change); 2 carry event time and a source key on every record; 3 make tasks idempotent and atomic; 4 write, audit, publish; 5 give each source a contract and diff the schema before loading; 6 layer dbt as staging, intermediate and marts; 7 maintain Iceberg tables on a schedule; 8 keep business history in rows (SCD Type 2); 9 pick the H3 resolution from the source and join with cell_to_parent; 10 define everything as code and make it cheap to destroy; 11 make pipelines observable; 13 least privilege for agents; 15 store every timestamp in UTC; 16 24/7 parts write only to always-on storage
Step 0.1 to 0.5, 1B.3, 1B.5, 1B.9	Steps of the platform skeleton and Project 1 in the build plan: 0.1 repo and CI, 0.2 cloud base (buckets, Glue, budget, roles), 0.3 session box, 0.4 serving template, 0.5 status and incidents; 1B.3 warehouse, 1B.5 AI foundations, 1B.9 operations hardening
Glue	AWS Glue Data Catalog: the catalog that lists the Iceberg lake tables so every engine reads the same tables
CloudFront	AWS service that delivers the website files
API Gateway	AWS service that receives web requests and passes them to Lambda; the HTTP type serves public pages, the REST type supports API keys
Terraform	The tool that creates the AWS resources from code
Partition	A slice of a table that can be rewritten on its own
Manifest	A list of files with their URL, size and hash, kept instead of a copy for public archives that never change
Red-team	Prompts written to make an agent misbehave, used as a test set

Term	Meaning
Serving template	The platform's shared pattern from step 0.4: publish gold tables as Parquet, answer with DuckDB in Lambda behind API Gateway, show on a React site
Marquez	The lineage graph tool of the platform (not used directly by Project 3 components)
ADR n	Architecture decision record n, in adrs/
OQ-n	Open question n, in design doc section 16
FR, QR, AC, QS	Functional requirement, quality requirement, acceptance criterion, quality scenario
Label	A recorded correct answer about a vessel pair, made by John; not to be confused with the label wording shown on the page

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1. Introduction and goals (arc42 section 1 style)

1.1 Requirements overview

Project 3 is the Ocean map. A visitor enters an ocean or coastal point and sees how much nearby seafloor is unmapped, maps and counts of vessel activity, and the latest buoy conditions with the time they were taken, with a cleaned vessel identity table behind it (build plan, Project 3 summary, reworded: vessel names and MMSI stay internal, ADR 0008).

The project is built in three phases, each ending deployed:

- Phase A (live): one region (for example the Gulf of Mexico), buoys, a seafloor grid, fishing activity and a public lookup page.
- Phase B (depth): a live vessel stream, buoy history and the Snowflake warehouse, vessel identity resolution with a human review queue, AI-assisted matching, a data-quality agent and buoy fault detection.
- Phase C (scale, second round): MarineCadastre's full 2024 AIS year, multi-node processing, global GEBCO coverage, vessel activity clusters and batch location briefings.

Phase C starts only after every project has reached Phase B (build plan, "How the plan works").

1.2 Quality goals

The four goals below are taken from recorded decisions, in this order of priority.

Priority	Quality goal	Why (decision)
1	Cost containment: nothing surprises with cost	Budget under $50 a month, a hard limit on every paid item (Build Plan Decisions, "Budget", "Surprise costs")
2	Availability with the session box off: the public lookup keeps answering when the session box is off	Public sites are static files plus Lambda; Lambda and Fargate write only to always-on storage (platform rule 16; decision "Always-on parts")
3	Honest data: labels never claim more than the data supports, and no AI change reaches the data without human approval	"possible dark vessels" (ADR 0004); mapped rule (ADR 0003); data-quality agent writes proposals only (ADR 0013)
4	Safe replay: reruns and replays write each event once	Idempotent tasks (platform rule 3); replaying a capture twice writes each event once (3B.1 Done when; ADR 0006, ADR 0007)

1.3 Stakeholders

Stakeholder	Expectation
John (owner and builder; the only person who decides review-queue items)	Builds the project from these documents. Approves each document before coding. Decides every match proposal.
Public visitor	Enters a point and gets an answer on the public link, with the session box off.

Stakeholder	Expectation
Reader of the portfolio (anyone assessing the work, including other engineers)	Can see the stack in use, the published numbers (lag, precision and recall, cost per 1,000 pairs) and the decision records.
Data providers (NOAA NDBC, GEBCO, Global Fishing Watch, AISStream)	Their licenses, terms and credit requirements are honored (ADR 0008, ADR 0009).

2. Users and needs

The build plan names no personas for Project 3 beyond the one-line summary. The users below are derived from that summary and from the Phase B steps that mention "you" (the builder).

User	Need	Served by
Visitor	Enter an ocean or coastal point and learn how much nearby seafloor is unmapped, how much fishing activity is nearby, what the nearest buoy reports, and how recent that is	3A.4 lookup page; features F-3A.1 to F-3A.4
Visitor	Understand that "possible dark vessels" are possible, not confirmed	3A.3 label wording
Visitor	Know which buoys are suspect	3B.6 buoy flags with the rule that fired
Maintainer (John)	Resolve an uncertain vessel identity match with one pair on one screen together with its evidence	3B.3 review queue
Maintainer (John)	See proposed fixes from the data-quality agent and approve or reject them, with nothing changing until approved	3B.5
Maintainer (John)	Know whether rules only, rules plus AI, or a trained classifier is the best matcher on measured precision, recall, cost and review volume	3B.4 comparison and decision record
Maintainer (John)	Know when a source changes format before the nightly run fails	Data contracts plus weekly smoke test (decision "Source changes")

3. Features by phase

IDs follow the build plan step numbers.

Phase A: live (scope: one region, for example the Gulf of Mexico; OQ-2)

ID	Feature	Build plan step
F-3A.1	Buoy readings: NOAA NDBC realtime data for the region's stations, with a contract and range checks on waves and temperature	3A.1
F-3A.2	Seafloor grid: GEBCO 2026 region cut, converted to H3 resolution 8, each cell flagged mapped or unmapped from GEBCO's source-type grid	3A.2
F-3A.3	Fishing activity: Global Fishing Watch 4Wings AIS fishing presence and SAR detections with no AIS match ("possible dark vessels"), stored as per-cell counts	3A.3
F-3A.4	Lookup page: dbt per-cell tables on DuckDB, published, shown on Project 1's map component with credits	3A.4

Phase B: depth

ID	Feature	Build plan step
F-3B.1	Live vessel stream: AISStream websocket through Kafka and Structured Streaming during sessions; daily one-hour Fargate capture to S3; nightly replay of captures through Kafka	3B.1
F-3B.2	Buoy history backfill by year, SCD Type 2 station metadata, marts on Snowflake	3B.2

ID	Feature	Build plan step
F-3B.3	Vessel identity resolution: golden vessel record, MMSI and IMO validation, MMSI collision detection, name variant matching, review queue, labels applied as SCD Type 2	3B.3
F-3B.4	AI-assisted matching: three versions compared (rules only; rules plus AI; trained classifier)	3B.4
F-3B.5	Data-quality agent that proposes fixes into the review queue and has no direct write access	3B.5
F-3B.6	Buoy fault detection: rules plus scikit-learn flag stuck, spiking or silent buoys	3B.6

Phase C: scale (second round)

ID	Feature	Source
F-3C.1	Process MarineCadastre's full 2024 AIS year (daily GeoParquet, CC0) month by month, measuring input size before claiming one, and recording rows, runtime per month and cost	build plan, Phase C: "Process MarineCadastre's full 2024 AIS year"
F-3C.2	Run one month on a 3 to 4 node spot cluster for "a few dollars" (the build plan's wording; the dollar limit is open, OQ-23) and record what changed versus one node	build plan, Phase C: "Run one month on a 3 to 4 node spot cluster for a few dollars"
F-3C.3	Snowflake reads the AIS-year Iceberg tables in place for the marts	build plan, Phase C: "Snowflake reads the AIS-year Iceberg tables in place for the marts"
F-3C.4	Global GEBCO coverage	build plan, Phase C: "Global GEBCO coverage"
F-3C.5	Vessel activity clusters with LLM labels marked as hypotheses	build plan, Phase C: "Vessel activity clusters with LLM labels marked as hypotheses"
F-3C.6	Batch location briefings that reject any number a query did not return	build plan, Phase C: "batch location briefings that reject any number a query did not return"
F-3C.7	Optional: run the identity-resolution code on a HubSpot free developer account seeded with clearly synthetic contacts, through its API and webhooks	build plan, Phase C: "Optional: run the identity-resolution code on a HubSpot free developer account"

The build plan gives no "Done when" checks for Phase C. They are missing and are recorded as missing; John decided (Oct 9, 2026) they are written when Phase C is designed (ADR 0019). The only stated gate is "the design note is updated with measured numbers" (Build order and checkpoints, second round).

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4. Functional requirements

Each requirement traces to a feature. "Source" cites the build plan step or the decision document.

Phase A

ID	Requirement	Feature	Source
FR-01	The Airflow DAG `ndbc_realtime_dag` (C1) pulls NOAA NDBC realtime data for the stations in the region. The station list source is identified in 3A.1 (OQ-7).	F-3A.1	3A.1
FR-02	The NDBC ingest (C1) loads every buoy reading with its load time, event time and a source key, and keeps the raw files immutable with source URL, fetch time and hash.	F-3A.1	3A.1; platform rules 1, 2
FR-03	A data contract exists for the NDBC source (see contracts/source-ndbc-realtime.yaml); the schema is diffed before loading and a change fails loudly.	F-3A.1	3A.1; platform rule 5

ID	Requirement	Feature	Source
FR-04	Range checks on wave values and temperature run in a Great Expectations publish gate; data failing the checks is not published. Range bounds are open (OQ-7).	F-3A.1	3A.1; platform rule 4
FR-05	The region is cut from the GEBCO 2026 grid with xarray or rasterio. The region is a bounding box fixed in 3A.2 (OQ-2).	F-3A.2	3A.2
FR-06	The cut is converted to H3 resolution 8 (ADR 0002).	F-3A.2	3A.2; platform rule 9
FR-07	GEBCO's source-type grid flags each ocean cell mapped or unmapped by the rule in ADR 0003. Codes 10 to 17 count as mapped. Codes 40 to 46 (interpolated or predicted) and 70 to 72 (unknown) do not count as mapped. Code 0 is land and is not an ocean cell. How the pixel codes inside one cell combine into a single cell flag is open (OQ-4).	F-3A.2	3A.2; Risks table "Mapped seafloor rule"
FR-08	The mapped/unmapped rule is written down in the dbt model description and in the design doc (section 7, C2).	F-3A.2	3A.2 ("Write the rule down")
FR-09	The Global Fishing Watch ingest (C3) requests 4Wings data for the region using an API key it reads from Parameter Store: AIS fishing presence (hours) and SAR detections.	F-3A.3	3A.3; platform "Key storage" decision
FR-10	SAR detections are filtered to those with no AIS match and labeled "possible dark vessels", never "dark vessels" without "possible" (ADR 0004).	F-3A.3	3A.3; Risks table "Dark vessels"
FR-11	The Global Fishing Watch ingest (C3) stores the gridded results as per-cell counts with load time (not per-vessel rows). The grid resolution and the H3 resolution used for storage are open (OQ-6).	F-3A.3	3A.3
FR-12	The ingest requests one 4Wings report at a time, and recovers reports that exceed 100 seconds through last-report (ADR 0009).	F-3A.3	Documentation Plan "Global Fishing Watch fields"
FR-13	The label wording ("possible dark vessels") and the Global Fishing Watch credit appear on the page. Exact credit text is open (OQ-24).	F-3A.3	3A.3
FR-14	dbt on DuckDB builds per-cell tables in the layers staging, intermediate and marts, with `source()` only in staging and `unique` and `not_null` tests on keys.	F-3A.4	3A.4; platform rule 6
FR-15	Gold tables are written, audited and then published as Parquet (write, audit, publish).	F-3A.4	3A.4; platform rule 4; skeleton step 0.4
FR-16	A lookup for a point returns the unmapped share of nearby seafloor, fishing activity and the nearest buoy's latest conditions, with credits for every source. The meaning of "nearby" and how a point is entered are open (OQ-5, OQ-9).	F-3A.4	3A.4
FR-17	The page uses Project 1's map component (the platform's React and TypeScript site template with a map component) (ADR 0005).	F-3A.4	3A.4; step 0.4
FR-18	The lookup is answered by Lambda running DuckDB over the published Parquet, with the session box off.	F-3A.4	3A.4 "Done when"; step 0.4
FR-19	The page states: "The first request after a period of inactivity may take up to 10 seconds while the service initializes."	F-3A.4	Step 0.4
FR-20 (proposed)	Every lookup answer carries an explicit "as of" time for the buoy readings and the fishing data, because the buoy source is refreshed by a daily run and not continuously.	F-3A.4	Proposed (needs John's approval, OQ-30); OQ-7
FR-59	When the nightly run is late or failed, the page shows the old data with a visible late notice and the time of the last good run. Data is never hidden. How "late" is determined and the notice wording are open (OQ-29).	F-3A.4	Decided by John, Oct 9, 2026 (ADR 0016); reason: hiding data would leave the visitor with nothing, while old data with a notice is still useful

Phase B

ID	Requirement	Feature	Source
FR-21	The AISStream websocket client runs only on the server side (session box or Fargate), never from a browser, and uses at most the three connections AISStream allows.	F-3B.1	Phase B first-run checks; Risks table "Live AIS terms"
FR-22	The AISStream key is read from Parameter Store.	F-3B.1	Decision "Key storage"
FR-23	During sessions, the live stream flows through Kafka and a Structured Streaming job into Iceberg.	F-3B.1	3B.1; decision "Live stream"
FR-24	Outside sessions, a daily one-hour Fargate capture writes raw messages to S3 only (Kafka is off then).	F-3B.1	3B.1; decision "Ship captures"
FR-25	The nightly run replays the day's captures through Kafka with the same streaming job using Spark's availableNow trigger, then stops.	F-3B.1	3B.1
FR-26	The job parses the semi-structured AISStream messages (MessageType, MetaData, typed Message), handles late and duplicate events and writes to Iceberg.	F-3B.1	3B.1; Documentation Plan "AISStream message format"
FR-27	A live session records lag (median and 95th percentile) and messages per second. The definition of lag is open (OQ-14).	F-3B.1	3B.1 Done when
FR-28	Replaying a capture twice writes each event once.	F-3B.1	3B.1 Done when
FR-29	Buoy history is backfilled by year past NDBC's 45-day realtime window; missing stations and years are logged. Backfills run only during work sessions (never in the nightly run).	F-3B.2	3B.2; decision "Backfills"
FR-30	Buoy station metadata is kept as SCD Type 2 rows.	F-3B.2	3B.2; platform rule 8
FR-31	The marts run on Snowflake (reading Iceberg through the Glue catalog, AWS's catalog that lists the lake tables), with a refresh step at the end of the pipeline run.	F-3B.2	3B.2; Risks table "Snowflake setup"
FR-32	MMSI values are validated against country prefixes. The prefix reference table is not in the build plan's source list (OQ-15; ADR 0017).	F-3B.3	3B.3
FR-33	IMO numbers are validated by check digit.	F-3B.3	3B.3
FR-34	One MMSI used by two ships at once is caught by impossible speed between positions. The speed limit is open (OQ-16).	F-3B.3	3B.3
FR-35	Name variants are matched. The method and thresholds are open (OQ-16).	F-3B.3	3B.3
FR-36	A golden vessel record is built from the matched observations.	F-3B.3	3B.3
FR-37	Low-confidence matches go to a review queue (DynamoDB and a React page, owner only, not linked from the public site).	F-3B.3	3B.3; platform table "Visitor data"
FR-38	The review handler (C15) stores John's decisions as labels (recorded correct answers, used to apply changes and to train and check matching) and applies them as SCD Type 2 rows (history kept as rows with valid-from and valid-to times).	F-3B.3	3B.3; platform rule 8
FR-39	The duplicate rate before and after is reported, with precision and recall on 200 hand-labeled pairs.	F-3B.3	3B.3 Done when
FR-40	Three matcher versions exist: rules only; rules plus AI (embeddings find candidate pairs, an LLM decides ambiguous ones through the batch API with a reason and confidence); and a small classifier trained on the LLM's decisions and John's labels, so the pipeline runs without model calls.	F-3B.4	3B.4
FR-41	Precision, recall, cost per 1,000 pairs and review-queue volume are reported for all three versions.	F-3B.4	3B.4 Done when
FR-42	The chosen version is recorded in a decision record (ADR 0012 is replaced or superseded by the result).	F-3B.4	3B.4 Done when

ID	Requirement	Feature	Source
FR-43	A data-quality agent reads failed checks and low-confidence matches and proposes fixes: merge two vessels, mark a buoy faulty, open an issue.	F-3B.5	3B.5
FR-44	The agent's only write tool creates a proposal in the review queue. Nothing changes until John approves.	F-3B.5	3B.5; platform rule 13
FR-45	Proposals on real cases are logged with John's decisions, and the approval rate is tracked.	F-3B.5	3B.5 Done when
FR-46	Rules plus scikit-learn flag stuck, spiking or silent buoys.	F-3B.6	3B.6
FR-47	The page shows the flag, the rule that fired and the readings, with no generated note (ADR 0014).	F-3B.6	3B.6; decision "Cut: LLM-written buoy notes"
FR-48	Every model call in Project 3 is logged to S3 (model, prompt version, tokens, latency, cost) through the platform's call log, using only the cheapest model and structured output.	F-3B.4, F-3B.5	Platform step 1B.5 (shared); decisions "AI spending", "Model"
FR-49	Every DAG writes freshness and row counts to the platform metrics table; a missed freshness target or failed run sends an email.	all	Step 0.5; platform rule 11
FR-57 (proposed)	The 200 hand-labeled evaluation pairs are never used to train the classifier or as examples in LLM prompts.	F-3B.3, F-3B.4	Proposed (needs John's approval, OQ-30; default recorded in ADR 0020); OQ-17
FR-58	Review-queue decisions can be written by the owner only, through the owner-only review routes: keyed REST type, one API key held in Parameter Store. The review page is not linked from the public site. The way the key reaches the page without being visible in browser code is open (OQ-31).	F-3B.3	Decided by John, Oct 9, 2026 (ADR 0015); reason: a write path on a public site needs access control, and one owner needs only one key

Note: FR-57, FR-58 and FR-59 were added after FR-56, so each is listed at the end of its own phase table.

Phase C

ID	Requirement	Feature	Source
FR-50	Input size of the AIS year is measured before any size is claimed; rows, runtime per month and cost are recorded.	F-3C.1	build plan, Phase C: "Process MarineCadastre's full 2024 AIS year"
FR-51	One month runs on a 3 to 4 node spot cluster, and the change versus one node is recorded.	F-3C.2	build plan, Phase C: "Run one month on a 3 to 4 node spot cluster for a few dollars"
FR-52	Snowflake marts read the AIS-year Iceberg tables in place.	F-3C.3	build plan, Phase C: "Snowflake reads the AIS-year Iceberg tables in place for the marts"
FR-53	GEBCO coverage becomes global.	F-3C.4	build plan, Phase C: "Global GEBCO coverage"
FR-54	Vessel activity clusters carry LLM labels explicitly marked as hypotheses.	F-3C.5	build plan, Phase C: "Vessel activity clusters with LLM labels marked as hypotheses"
FR-55	Batch location briefings reject any number that a query did not return.	F-3C.6	build plan, Phase C: "batch location briefings that reject any number a query did not return"
FR-56	Optional: the identity-resolution code runs on a HubSpot free developer account seeded with clearly synthetic contacts, through its API and webhooks.	F-3C.7	build plan, Phase C: "Optional: run the identity-resolution code on a HubSpot free developer account"

5. Quality requirements

Each quality requirement has a verification method. Numeric thresholds appear only where a decision or confirmed fact sets them; elsewhere the threshold is set when the measurement exists and the question is listed (OQ-n).

ID	Attribute	Requirement	Threshold / source	How verified
QR-01	Cost	Total platform spend stays under the monthly budget	Under $50 a month (decision "Budget")	Cost per run and monthly total in Operations docs; AWS Budget of $40 as backup (platform step 0.2)
QR-02	Cost	Each Spark job has a maximum rows read and written and a maximum runtime; a job over either limit stops before publishing and opens an incident	Limits set from measured sizes (OQ-1, OQ-23)	CI test with an oversized sample file confirms the job stops before publish; production incident record
QR-03	Cost	Model spend stays within the monthly limit with the cheapest model only	$15 monthly spend limit set in the API account before the first call; daily question cap and token limit per answer (decision "AI spending")	Call log loaded into the llm_calls table; spend limit visible in the API account
QR-04	Cost	Sessions and the nightly run are time-boxed	4-hour session limit; nightly run hard maximum of 2 hours; hitting the maximum is recorded as an incident	Platform guards (build plan step 0.3, the session box step; documented in the platform design doc and platform operations); incident record in operations.md
QR-05	Availability	The public lookup answers with the session box off	Done-when check of 3A.4	Manual check with the session box off after each deploy; smoke test in CI against the dev deploy
QR-06	Latency	Cold-start behavior is stated honestly on the page	Page text: first request after inactivity may take up to 10 seconds (step 0.4); median warm latency and 95th-percentile cold-start time reported in the README (step 1B.9, shared)	Measurement script from the platform; README figures
QR-07	Freshness	The nightly run refreshes every pipeline once a day	Nightly start 10:00 UTC (default used, decision "Nightly run time"); numeric freshness target per table is open (OQ-8)	Freshness and row counts per run in the metrics table; status page shows last good run
QR-08	Freshness	A failed or late nightly run never hides data: the page shows the old data with a visible late notice and the last good run time, and every answer carries an "as of" time	Late-data rule: Decided by John, Oct 9, 2026 (ADR 0016); the "as of" time is proposed (FR-20, OQ-30); lateness threshold open (OQ-29)	Test in CI with an aged sample: data still shown, notice shown, last good run time shown
QR-09	Correctness	Replaying a capture twice writes each event once	3B.1 Done when	CI test: replay a sample capture twice and compare row counts and keys
QR-10	Correctness	Reruns of any pipeline task are safe	Overwrite a partition (a slice of a table that can be rewritten on its own); use the run's data interval, not now() (platform rule 3)	CI test: run a task twice on a sample and compare output
QR-11	Accuracy	Vessel identity matching meets published precision and recall	Reported on 200 hand-labeled pairs; no pass threshold is set by the plan (OQ-17)	Evaluation job against the labeled set, reported for all three versions (3B.3, 3B.4)

ID	Attribute	Requirement	Threshold / source	How verified
QR-12	Accuracy	The mapped-seafloor flag follows GEBCO's source-type codes exactly	Codes 10 to 17 mapped; 40 to 46 interpolated or predicted; 70 to 72 unknown; 0 land	Unit test with a sample tile of known codes; first-run check "GEBCO's source-type codes are read correctly for the region"
QR-13	Data integrity	Raw files are immutable with source URL, fetch time and hash; live raw files are locked 30 days; public archives that never change (GEBCO) are kept as a manifest (a list of files with their URL, size and hash)	Platform rule 1; decision "Raw-file lock"	Object Lock configuration check; manifest present
QR-14	Data integrity	Each source has a contract; the schema is diffed before loading; a weekly smoke test pulls one small sample from every source	Platform rule 5; decision "Source changes"	Contract tests in CI; weekly smoke test result; first-run checks
QR-15	Data integrity	Bad data never reaches the site	Write, audit, publish (platform rule 4)	Great Expectations gate test in CI: a failing sample stops publishing
QR-16	Security	All keys (Global Fishing Watch, AISStream, model API) live in Parameter Store and none in code or GitHub; each pipeline has a least-privilege role; dev roles cannot write live data	Platform rules 10, 13; step 0.2	gitleaks scan in CI; IAM policy review; refused-write check from the dev role
QR-17	Security	The data-quality agent has no direct write access; its only write is a proposal in the review queue	3B.5 Done when	IAM policy test: agent role cannot write to lake or contract tables; red-team prompts (prompts written to make the agent misbehave) including instructions hidden in source text (platform rule 13)
QR-18	Security	The AISStream key is never exposed to a browser	Phase B first-run check	Static check that no client bundle contains the key; connection originates only from the session box or Fargate
QR-19	Privacy	No personal data is collected from visitors by this project	The lookup has no write path in the build plan	Design review; no visitor write endpoint in openapi.yaml
QR-20	Privacy	Vessel identifiers (MMSI, ship names) are never shown publicly; only maps and counts with credit are shown (ADR 0008, decided by John, Oct 9, 2026). The rule is loosened only after AISStream answers John's question about their terms (OQ-3)	Decision of Oct 9, 2026	Page review before each release; contract `limitations` text; check that no public response contains a vessel name or MMSI
QR-21	Time	Every timestamp is stored in UTC and converted only on the page	Platform rule 15	dbt test that timestamp columns carry UTC; page test of the conversion
QR-22	Observability	Freshness and row counts per run, a public status page, one ticket per incident	Platform rule 11; step 0.5	Status page shows each pipeline's last good run
QR-23	Observability	A live session records lag (median and 95th percentile) and messages per second	3B.1 Done when	Metrics table row per session

ID	Attribute	Requirement	Threshold / source	How verified
QR-24	Legal	Non-commercial and share-alike licenses are honored	Global Fishing Watch non-commercial; credit on page and in README (assumptions paragraph of the plan)	README and page review; licenses and attribution register in security-and-privacy.md
QR-25	Explainability	Buoy flags show their rule and readings; no generated text	3B.6 Done when; decision "Cut"	Page test; no model call in the flag path (call log shows none)
QR-26 (proposed)	Evaluation	Agent and matching evals run against known answers	Technology table lists "LLM evals, LLM-as-judge, red-team, tracing" for Project 3 phase B; the proposed step list in section 8 defines them, awaiting John's approval (OQ-21)	Eval harness in CI once the steps are approved
QR-27	Documentation	Every function in the code carries a docstring; reference docs are generated	Documentation Plan "Reference, generated from code"	pdoc or Sphinx build in CI

5.1 Quality scenarios (arc42 section 10 style)

ID	Scenario	Stimulus	Environment	Expected response	Measure
QS-01	A visitor looks up a point while no session box is running	HTTP request from the page	Box off, Lambda cold	Lookup answered from published Parquet by DuckDB in Lambda	Answer returned; page shows the cold-start notice; latency reported in the README
QS-02	The nightly run fails halfway	Spark job error	Nightly run at 10:00 UTC	Publish gate stops; previous published data remains; status page shows last good run; an email goes out and an incident record is written	No new publish for that night; email sent; incident entry exists
QS-03	Spot is unavailable at 10:00 UTC	Spot capacity refused	Nightly run	The night is skipped; the next run catches up safely; daily captures remain in S3 and replay at the next run	Skipped run recorded; next run replays both days' captures with each event written once
QS-04	A capture is replayed twice	Rerun of the replay task	Box on	Same rows	Row count and keys identical (QR-09)
QS-05	NDBC changes its file format	Upstream change	Weekly smoke test	Contract schema diff fails loudly before the nightly run	Smoke test alert before the next nightly run
QS-06	The data-quality agent proposes a merge of two vessels	Agent run	Agent on Lambda or in a pipeline (OQ-19)	A proposal appears in the review queue; no table changes	Zero writes to lake tables from the agent role; proposal logged
QS-07	Source text contains an instruction aimed at the agent (for example in a vessel name)	Planted text in a name field	Red-team set	The agent treats it as data and does not act on it	Red-team cases pass (threshold set when the proposed eval steps in section 8 are approved)
QS-08	A model-call bill climbs	Many LLM matching calls	Batch API in Phase B	Spend stops at the monthly limit	$15 monthly limit not exceeded; call log shows cost per call
QS-09	A Global Fishing Watch report takes longer than 100 seconds	Slow 4Wings report	Nightly or session run	Report recovered via last-report	Report eventually stored; no second concurrent report started (ADR 0009)
QS-10	A Spark job reads more rows than its limit	Unexpectedly large input	Any Spark job	Job stops before publishing and opens an incident	No publish; incident opened

6. Acceptance criteria

One block per feature. Each criterion is the build plan's "Done when" check (quoted from the plan, split into testable parts). Phase C done-when checks are missing and are written when Phase C is designed (decided by John, Oct 9, 2026).

F-3A.1 Buoys (3A.1)

- AC-3A.1-1: Readings land with load time.
- AC-3A.1-2: The check tests pass in CI.

F-3A.2 Seafloor grid (3A.2)

- AC-3A.2-1: Every ocean cell in the region has a mapped flag.
- Supporting check (from the task text): the mapped/unmapped rule is written down.

F-3A.3 Fishing activity (3A.3)

- AC-3A.3-1: Per-cell counts land with load time.
- AC-3A.3-2: The label wording is on the page.

F-3A.4 Lookup page (3A.4)

- AC-3A.4-1: The public link answers a lookup with the session box off.
- AC-3A.4-2 (proposed, follows FR-20; OQ-30): Every lookup answer states when its buoy and fishing data was taken (an "as of" time, shown in local time). Exact wording is pending OQ-29.
- AC-3A.4-3: When the nightly run is late or failed, the lookup still returns the old data, with a visible late notice and the time of the last good run (decided by John, Oct 9, 2026; ADR 0016).
- Supporting checks (from the task text): a point shows unmapped share, fishing activity and the nearest buoy's latest conditions, with credits; the page reuses Project 1's map.

F-3B.1 Live vessel stream (3B.1)

- AC-3B.1-1: A live session records lag (median and 95th percentile) and messages per second.
- AC-3B.1-2: The nightly replay processes the day's capture and stops.
- AC-3B.1-3: Replaying a capture twice writes each event once.

F-3B.2 Buoy history and warehouse (3B.2)

- AC-3B.2-1: The multi-year backfill runs in Airflow.
- AC-3B.2-2: Gaps are logged.
- AC-3B.2-3: The marts pass tests on Snowflake.

F-3B.3 Vessel identity resolution (3B.3)

- AC-3B.3-1: The duplicate rate before and after is reported.
- AC-3B.3-2: Precision and recall on 200 hand-labeled pairs are reported.

F-3B.4 AI-assisted matching (3B.4)

- AC-3B.4-1: Precision, recall, cost per 1,000 pairs and review-queue volume are reported for all three versions.
- AC-3B.4-2: The chosen version is in a decision record.

F-3B.5 Data-quality agent (3B.5)

- AC-3B.5-1: Proposals on real cases are logged with John's decisions.
- AC-3B.5-2: The approval rate is tracked.
- AC-3B.5-3: The agent has no direct write access.

F-3B.6 Buoy fault detection (3B.6)

- AC-3B.6-1: Flags appear on the page with their rule.
- AC-3B.6-2: The check tests pass in CI.

Phase gates (Build order and checkpoints)

- Gate for Project 3, Phases A and B: the ocean lookup is public; the live stream, identity resolution and review queue work.
- Gate for Phase C: the design note is updated with measured numbers.

---

7. Constraints from the platform (applied, not repeated)

These come from the platform documents (platform design doc, platform ADRs) and from the plan's engineering standards. They are listed so each can be traced to a Project 3 component.

Rule	Applies in Project 3 to
1 Raw immutable, manifest for static archives	NDBC raw files, AISStream captures, GFW responses; GEBCO as manifest
2 Event time and source key on every record	Buoy readings, AIS events, GFW cells
3 Idempotent, atomic tasks	Every DAG and Spark job; replay of captures
4 Write, audit, publish	Publish of lookup tables; Snowflake marts
5 Contracts and schema diff	contracts/
6 dbt layers and tests	Per-cell models, marts
7 Iceberg maintenance as last nightly task	All Iceberg tables in this project
8 Business history in rows (SCD Type 2)	Station metadata, vessel golden record
9 H3 resolution from the source; join with cell_to_parent	Seafloor cells at 8 (ADR 0002); fishing cells at the resolution set by OQ-6
11 Observability	Metrics table rows, status page
13 Least privilege for agents	Data-quality agent (ADR 0013)
15 UTC everywhere, convert on the page	All tables and the lookup page
16 24/7 parts write only to always-on storage	Fargate capture writes to S3 only

---

8. Proposed step list for metrics, evals, tracing and MCP (PROPOSED, for John's approval)

The technology table lists MetricFlow, MCP servers and agents, and LLM evals, red-team and tracing for Project 3 Phase B, but no step defines them. John decided (Oct 9, 2026) that a short proposed step list is written. Nothing here is approved or built (ADR 0018). Numbers are not set; sizes marked "mirrors Project 1" copy the wording of Project 1 steps 1B.5 and 1B.7 and need approval.

Proposed step	Tasks	Proposed done when
3B.7 Metrics (proposed)	Define a small set of MetricFlow metrics on the Snowflake marts. Candidates are quantities this plan already names: duplicate rate, review-queue volume, approval rate of agent proposals, and unmapped share. The final set is John's choice.	One metric gives the same value from Snowflake and from DuckDB (mirrors step 1B.3).
3B.8 Call log and tracing (proposed)	Use the platform call log (model, prompt version, tokens, latency, cost) for the batch matching calls and the data-quality agent. Trace every agent run to AWS X-Ray.	A test agent run shows as a trace with its cost.
3B.9 MCP server (proposed)	Expose the metrics as read-only tools. Raw SQL is a read-only, allowlisted fallback logged separately. The data-quality agent reads through it. The agent's one write remains `create_proposal` (ADR 0013).	A test question runs end to end through the metrics tool.

Proposed step	Tasks	Proposed done when
3B.10 Evals in CI (proposed)	A question set with known answers for the agent, including unanswerable ones, graded exactly on tool choice, numbers and refusal. A red-team set covering instructions hidden in source text (for example a vessel name), write requests, off-topic requests and cost abuse. Sizes and thresholds are set when approved (mirrors Project 1: 30 or more questions, 20 or more red-team prompts).	CI fails below the accuracy and red-team thresholds.

Proposed additions (need John's approval)

These items are not John decisions and are listed in the design doc as OQ-30 and OQ-21:

1. FR-20 (proposed): an explicit "as of" time on every lookup answer.
2. FR-57 (proposed): the 200 evaluation pairs are never used for training or in prompts.
3. QR-26 (proposed) and the step list in section 8: metrics, evals, tracing and MCP for Project 3.

Part 3: Design document

`Ocean Map/design-doc.md`

Project 3: Ocean map. Design doc

Template: Google design doc, arc42, C4 (Documentation Plan). Seventeen sections in the order of the template.

---

1. Status, owner, reviewers, last updated

Field	Value
Status	Draft, awaiting John's review and approval. Coding of Project 3 does not start until approved.
Owner	John
Reviewers	John
Last updated	Oct 9, 2026
Build order position	Second project, after Project 1 (Documentation Plan: Project 1, Project 3, Project 5, Project 2, Project 4)
Companion documents	requirements.md, adrs/, contracts/, openapi.yaml
Platform documents	platform design doc, platform ADRs, platform contracts, platform requirements (written first; draft awaiting John's approval)

How to read value status. Every value in this document is either a recorded decision (cited as ADR n, see section 14) or a confirmed fact from the Documentation Plan, the Build Plan Decisions or the Build Plans for the Five Projects. Names of functions, classes, tables and fields are proposed here so the parts can be built. They are not recorded decisions, and they may be renamed in code if this document is updated to match. Everything not yet determined is in section 16. "Platform rule n" means engineering standard n of the build plan; the rules used by Project 3 are listed with their content in requirements.md, "Terms and abbreviations".

---

2. Summary

Ocean map lets a visitor enter an ocean or coastal point in one region (for example the Gulf of Mexico) and see how much nearby seafloor is unmapped (from GEBCO's source-type grid on an H3 resolution 8 grid), how much fishing activity is nearby (from Global Fishing Watch, including SAR detections with no AIS match labeled "possible dark vessels"), as maps and counts of vessel activity, and the nearest NOAA NDBC buoy's latest conditions with the time they were taken, all with credits. Phase A publishes this as a static-plus-Lambda lookup that works with the session box off. Phase B adds a live vessel stream from AISStream through Kafka and Structured Streaming (with a daily one-hour Fargate capture replayed nightly), buoy history in Snowflake marts, a cleaned vessel identity table built by rules, AI-assisted matching and a human review queue, a data-quality agent that can only propose fixes, and buoy fault detection with rules and scikit-learn. Phase C scales to MarineCadastre's full 2024 AIS year on a multi-node spot cluster and to global GEBCO coverage.

---

3. Context, scope and constraints

3.1 Background facts

Fact	Value	Source
Seafloor grid	GEBCO 2026 grid, with a source-type identifier grid. Codes 10 to 17 are direct measurements (echo sounders, seismic, lidar, charts, optical); 40 to 46 interpolated or predicted; 70 to 72 unknown; 0 land.	Build plan risks table; Documentation Plan "Which GEBCO codes mean measured"

Fact	Value	Source
Fishing activity	Global Fishing Watch 4Wings report endpoint returns CSV or JSON with date, lat, lon and hours or detections; SAR detections are filterable to matched or unmatched; one report at a time per user; reports over 100 seconds are recovered through last-report. API key required; credit required; non-commercial.	Documentation Plan; 3A.3
Dark vessels	SAR detections with no AIS match are possible dark vessels, not confirmed ones.	Risks table "Dark vessels"
Live AIS	AISStream.io: free key, three connections, must not be opened from a browser, no terms page. Subscribe with an API key, bounding boxes and optional MMSI and message-type filters; each message has MessageType, MetaData (MMSI, ship name, position) and the typed Message; 25 message types. Message rate for the region is not yet measured.	Phase B first-run check; Risks table; Documentation Plan
Buoys	NOAA NDBC realtime data (45-day realtime window); history by year past that window.	3A.1; 3B.2
Phase C AIS	MarineCadastre full 2024 AIS year, daily GeoParquet, CC0.	Build plan, Phase C: "Process MarineCadastre's full 2024 AIS year"
H3 resolutions	8 for 15 arc-second rasters; join across sources with cell_to_parent.	Platform rule 9

3.2 System boundary

In scope: ingest of NDBC, GEBCO, Global Fishing Watch and AISStream; transformations; published lookup tables; the Lambda lookup endpoint; the lookup page; vessel identity resolution; the review queue and its page; the data-quality agent; buoy fault detection.

Outside the boundary (owned by the platform; documented in `../Platform/`): Terraform bootstrap (Terraform is the tool that creates the AWS resources from code), the S3 buckets and the Glue catalog (AWS's catalog that lists the lake tables), CloudFront (AWS's service that delivers the website files), CI (automatic tests on every change), the session box and nightly schedule, Airflow and Spark installation, Kafka installation, Parameter Store, the serving template, the status page and metrics table, the AI foundations (call log, tracing, eval harness), Snowflake account setup and the refresh step, Marquez and the dbt docs site.

3.3 Constraints

Constraint	Value	Source
Budget	Under $50 a month, browser only; hard limit on every paid item	Decisions "Budget", "Surprise costs"
AI model use	Cheapest model only for now; $15 monthly spend limit; batch calls and cached content; daily question cap; token limit per answer	Decisions "AI spending", "Model"
Licensing	Global Fishing Watch is non-commercial; show only derived maps and counts of AIS data with credit	Plan assumptions; Risks table
Privacy	The lookup collects no visitor data. The review queue holds vessel data and John's decisions.	No visitor write path in the plan
Always-on rule	Lambda and Fargate write only to S3, DynamoDB, CloudWatch and X-Ray	Platform rule 16
Environments	Dev and live are separate in storage, catalogs and permissions	Decisions "Dev and live"
Time	All timestamps in UTC, converted on the page	Platform rule 15
Work method	Walking skeleton, small merges, CI with Spark on sample files, dev-first deploy	"How the work is done"
Spot only	Nightly run uses spot only; if unavailable, the night is skipped and the next run catches up	Nightly run decision

3.4 Decisions by John, Oct 9, 2026 (answered open questions)

Decision	Detail	ADR
Late or failed nightly run	The page shows the old data with a visible late notice and the time of the last good run. Data is never hidden. The "as of" requirement stays.	0016
Review queue writes	Owner only, behind one API key held in Parameter Store. The page is not linked from the public site. Keyed REST type for those routes.	0015
AISStream display	Maps and counts only; vessel names and MMSI internal. John will ask AISStream about their terms; the rule is loosened only after their answer.	0008
MMSI prefix reference	Stays an open question until a published reference table is chosen. The prefix list is not guessed.	ADR 0017 (OQ-15 stays open)
Classifier training	Databricks Free Edition with manual export (download results and model, commit to GitHub), as Projects 1 and 5. Whether export can be automated stays an open test.	0012
Metrics, evals, tracing, MCP	A short proposed step list, for John's approval (requirements.md section 8).	ADR 0018 (proposed; OQ-21)
Phase C done-when checks	Written when Phase C is designed; recorded as missing until then.	ADR 0019
Evaluation pairs	The 200 hand-labeled pairs are never used to train the classifier or as prompt examples. Approved by John, Oct 9, 2026.	ADR 0020 (accepted; OQ-30 covers the other proposals)
Buoy wording	Say "latest buoy reading" with its time, not "current conditions".	ADR 0021 (default)

4. Goals and non-goals

Goals

- Phase A: a public lookup for one region showing unmapped seafloor share, fishing activity and nearest-buoy conditions, working with the session box off.
- Phase B: a live and replayable vessel stream; buoy history; Snowflake marts; a measured vessel identity table with published precision and recall; a measured comparison of three matching versions; a data-quality agent limited to proposals; buoy fault flags with their rules.
- Phase C: measured scale numbers (input size, rows, runtime per month, cost; one-node versus 3 to 4 node).

Non-goals

- Confirming that a SAR detection is a dark vessel. The wording stays "possible dark vessels" (ADR 0004).
- Showing vessel names, MMSI or individual tracks publicly. Only maps and counts with credit are shown; vessel names and MMSI stay internal (ADR 0008, decided by John, Oct 9, 2026).
- LLM-written buoy notes. Cut by decision (ADR 0014).
- Any AI change to data without approval (ADR 0013).
- Collecting visitor reports, votes or sign-ups in this project. The build plan defines none.
- Commercial use. Licensing must be checked before charging for any project (plan assumptions).
- Opening the AISStream connection from a browser (ADR 0008).
- Per-vessel rows from Global Fishing Watch. The data is gridded; per-cell counts are stored (3A.3).
- Staged failure tests. Real incidents are recorded instead (decision "Failure testing").

---

5. Requirements

Full text in requirements.md. Summary:

Group	Summary	Requirement IDs
Phase A functional	NDBC buoys with contract and range checks; GEBCO region cut to H3 res 8 with mapped flag; GFW per-cell counts with "possible dark vessels" label; dbt per-cell tables on DuckDB, published, shown on Project 1's map, answered with the session box off	FR-01 to FR-20
Phase B functional	AISStream through Kafka with availableNow replay of daily captures; buoy history and SCD Type 2 station metadata; Snowflake marts; vessel identity golden record with review queue; three-version matching comparison; proposal-only agent; buoy fault flags	FR-21 to FR-49, FR-57, FR-58
Phase C functional	AIS year month by month, multi-node run, Snowflake reads in place, global GEBCO, hypothesis-labeled clusters, number-checked briefings, optional HubSpot demo	FR-50 to FR-56
Quality	Cost, availability with session box off, freshness, correctness of replay, accuracy of matching, security of keys and agent, privacy, time handling, observability, legal credit	QR-01 to QR-27, scenarios QS-01 to QS-10
Acceptance	One block per feature, quoting the build plan's "Done when"	requirements.md section 6

---

6. Architecture

6.1 System context (C4 level 1)

```
flowchart LR
    visitor(["`Visitor
enters a point`"]):::person
    maintainer(["`Maintainer (John)
approves or rejects proposals`"]):::person
    ocean["`Ocean map
lookup page, pipelines,
vessel identity, review queue`"]:::sys
    ndbc["`NOAA NDBC
buoy realtime and history`"]:::ext
    gebco["`GEBCO 2026 grid
depth and source type`"]:::ext
    gfw["`Global Fishing Watch
4Wings API`"]:::ext
    ais["`AISStream.io
live AIS websocket`"]:::ext
    model["`Model API
batch calls`"]:::ext
    snow["`Snowflake
warehouse for marts`"]:::ext
    visitor -->|"looks up a point, HTTPS"| ocean
    maintainer -->|"reviews proposals, HTTPS"| ocean
    ocean -->|"pulls readings and history"| ndbc
    ocean -->|"reads region cut"| gebco
    ocean -->|"4Wings reports, API key"| gfw
    ocean -->|"websocket, API key, server side only"| ais
    ocean -->|"batch matching calls"| model
    ocean -->|"marts via Iceberg and Glue"| snow
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#d1fae5,stroke:#047857,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef sys fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

6.2 Containers (C4 level 2)

Containers are split in two views, the always-on parts and the session box. Platform containers (Airflow, Spark, Kafka, k3s, S3, Glue, CloudFront, API Gateway (AWS's front door for web requests), Parameter Store, DynamoDB, X-Ray) are described in the platform design doc and shown here only where Project 3 uses them.

```
flowchart LR
    visitor(["Visitor"]):::person
    maintainer(["Maintainer (John)"]):::person
    ext["AISStream"]:::ext
    subgraph on["Always on (session box off)"]
        direction LR
        site["`Lookup page
React, CloudFront, S3
Project 1 map`"]:::cont
        lookup["`Lookup API
Lambda, FastAPI, DuckDB`"]:::cont
        parquet[("`Published Parquet
ocean_cell_summary
ocean_buoy_latest`")]:::store
        review["`Review page and API
React, Lambda
owner only, one API key,
keyed REST type,
not linked from public site`"]:::cont
        queue[("`Review queue
DynamoDB, point-in-time recovery on`")]:::store
        capture["`AIS capture
Fargate, 1 hour a day`"]:::cont
        rawS3[("`Raw files and captures
S3, versioned,
Object Lock 30 days`")]:::store
    end
    visitor -->|"uses"| site
    site -->|"GET lookup"| lookup
    lookup -->|"DuckDB reads"| parquet
    maintainer -->|"decides"| review
    review -->|"reads, writes decisions"| queue
    capture -->|"websocket"| ext
    capture -->|"writes raw parts"| rawS3
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#d1fae5,stroke:#047857,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef sys fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

The session box containers, with the always-on stores they touch shown outside the session box.

```
flowchart LR
    ext["`NDBC, GEBCO,
Global Fishing Watch`"]:::ext
    snow["Snowflake"]:::ext
    rawS3[("`Raw files and captures
S3`")]:::store
    parquet[("`Published Parquet
S3`")]:::store
    queue[("`Review queue
DynamoDB`")]:::store
    subgraph box["Session box (k3s, spot): sessions and nightly 10:00 UTC"]
        direction LR
        airflow["`Airflow DAGs
ingest, transform,
publish, backfill`"]:::cont
        spark["`Spark jobs
GEBCO to H3, AIS stream,
identity, fault detection`"]:::cont
        kafka["`Kafka
one broker`"]:::cont
        dbt["`dbt
DuckDB and Snowflake targets`"]:::cont
        ge["`Great Expectations
publish gate`"]:::cont
        agent["`Data-quality agent
Model API, MCP
location open: OQ-19`"]:::cont
        lake[("`Iceberg lake tables
S3 plus Glue`")]:::store
    end
    airflow -->|"pulls NDBC, GEBCO, GFW"| ext
    airflow -->|"writes raw files"| rawS3
    airflow -->|"runs"| spark
```

```
    spark -->|"produces and consumes"| kafka
    spark -->|"writes Iceberg"| lake
    dbt -->|"builds models"| lake
    dbt -->|"gold tables"| ge
    ge -->|"gates publish"| parquet
    agent -->|"creates proposals"| queue
    lake -->|"read in place"| snow
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#d1fae5,stroke:#047857,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef sys fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

6.3 Components (C4 level 3)

Component details are in section 7. First the Phase A pipeline and the buoy history path.

```
flowchart LR
    c1["`C1 NDBC ingest
Airflow DAG`"]:::cont
    c2["`C2 Seafloor grid
Spark, xarray or rasterio`"]:::cont
    c3["`C3 Fishing activity ingest
Airflow DAG`"]:::cont
    c12["`C12 Buoy history backfill
Airflow DAG`"]:::cont
    c18["`C18 Buoy fault detection
rules, scikit-learn`"]:::cont
    subgraph dbtg["Transform and publish"]
        direction LR
        c4["`C4 Ocean dbt project
dbt`"]:::cont
        c5["`C5 Publish gate
Great Expectations`"]:::cont
        c6["`C6 Publisher
Parquet writer`"]:::cont
    end
    c1 -->|"raw readings"| c4
    c2 -->|"cell flags"| c4
    c3 -->|"cell counts"| c4
    c4 -->|"gold tables"| c5
    c5 -->|"pass"| c6
    c12 -->|"history"| c18
    c18 -->|"flags"| c4
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#d1fae5,stroke:#047857,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef sys fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

Phase B streaming, identity and review components.

```
flowchart LR
    c9["`C9 AIS capture
Fargate`"]:::cont
    c10["`C10 AIS live producer
Kafka producer`"]:::cont
    c11["`C11 AIS stream job
Structured Streaming`"]:::cont
    c14["`C14 Vessel identity
Spark and Python`"]:::cont
    c16["`C16 AI matching
LanceDB, batch API, scikit-learn`"]:::cont
    c15["`C15 Review queue
DynamoDB, Lambda, React`"]:::cont
    c17["`C17 Data-quality agent
Model API, MCP`"]:::cont
    c9 -->|"replay via Kafka"| c11
    c10 -->|"live via Kafka"| c11
    c11 -->|"ais_messages"| c14
    c16 -->|"candidate pairs and decisions"| c14
    c14 -->|"low-confidence pairs"| c15
    c17 -->|"proposals"| c15
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
```

```
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#d1fae5,stroke:#047857,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef sys fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

---

7. Component breakdown

Naming note: identifiers are proposed at design time (see section 1). Project 3 code lives in the repo's `pipelines/`, `dbt/`, `checks/` and `app/` folders (step 0.1); the exact subfolder names are set at build.

Phase A components

C1 NDBC ingest (3A.1)

- Job: pull NOAA NDBC realtime data for the region's stations, keep raw files, load readings.
- Inputs: NDBC realtime files (contract source-ndbc-realtime); the station list for the region (source-ndbc-stations, OQ-7); the DAG run's data interval.
- Outputs: immutable raw files with source URL, fetch time and hash; staged readings; Iceberg `buoy_readings` (table-buoy-readings); freshness and row-count metrics.
- Main functions and classes: `ndbc_realtime_dag` (Airflow DAG); `list_region_stations(region)`; `fetch_station_file(station_id)`; `parse_station_file(raw_bytes)`; `check_contract_schema(parsed)`; `load_readings(parsed, interval)`.

C2 Seafloor grid (3A.2)

- Job: cut the region from the GEBCO 2026 grid, convert to H3 resolution 8 and flag each ocean cell mapped or unmapped.
- Inputs: GEBCO grid and source-type grid (source-gebco-grid); region bounding box (OQ-2).
- Outputs: Iceberg `seafloor_cells` (table-seafloor-cells); a manifest (a list of files with their URL, size and hash) for the GEBCO archive (platform rule 1: keep raw files immutable with source URL, fetch time and hash, and for public archives that never change store a manifest instead of a full copy).
- Main functions and classes: `cut_region(grid_path, bbox)` (xarray or rasterio); `pixels_to_h3(pixels, resolution=8)`; `classify_source_code(code)`; `aggregate_cell_flag(pixel_codes)`; `build_seafloor_cells(region)` (Spark job).
- The rule (ADR 0003): codes 10 to 17 count as mapped; 40 to 46 and 70 to 72 do not; 0 is land and the cell is not an ocean cell. How many pixel codes decide a cell flag, and how the unmapped share is derived, is OQ-4. The rule is also written in the dbt model description.

C3 Fishing activity ingest (3A.3)

- Job: request Global Fishing Watch 4Wings reports for the region and store per-cell counts.
- Inputs: Global Fishing Watch API key from Parameter Store; region; date interval from the run; datasets and temporal resolution (OQ-6).
- Outputs: raw responses with hash and load time; Iceberg `fishing_cells` (table-fishing-cells) with AIS fishing hours and SAR unmatched detections ("possible dark vessels").
- Main functions and classes: `gfw_fishing_dag` (Airflow DAG, concurrency of one GFW request at a time, ADR 0009); `request_4wings_report(params)`; `recover_last_report(session)` (for reports over 100 seconds); `parse_report(csv_or_json)`; `to_h3_cells(rows, resolution)`; `load_fishing_cells(rows, interval)`.

C4 Ocean dbt project (3A.4, 3B.2)

- Job: staging, intermediate and mart models; per-cell tables; tests.
- Inputs: Iceberg tables from C1, C2, C3 (Phase A); also buoy history, stations, flags, AIS and vessel tables (Phase B).
- Outputs: `ocean_cell_summary` and `ocean_buoy_latest` (table-ocean-cell-summary, table-ocean-buoy-latest); Snowflake marts in Phase B.

- Main models (layers per platform rule 6): `stg_buoy_readings`, `stg_seafloor_cells`, `stg_fishing_cells`, `int_cell_summary`, `int_nearest_station`, `mart_ocean_cell_summary`, `mart_ocean_buoy_latest`. Tests: `unique` and `not_null` on keys; accepted values on mapped flag.
- Targets: DuckDB (development, CI, Phase A publish) and Snowflake (production marts, Phase B). The same models run in both (Tool table "Transformations").

C5 Publish gate (3A.1, 3A.4)

- Job: Great Expectations checks on freshness, empty results and value ranges, including range checks on waves and temperature.
- Inputs: staged tables before publish; contracts.
- Outputs: pass or fail; a failure stops publishing and records a check result row.
- Main functions and classes: `ocean_publish_suite` (expectation suite); `run_gate(table, suite)`; `record_gate_result(result)`. Range bounds are OQ-7.

C6 Publisher (3A.4)

- Job: write gold tables as Parquet after the gate passes (write, audit, publish).
- Inputs: gold tables from C4.
- Outputs: Parquet under the published location read by C7.
- Main functions: `write_staging_parquet(table)`; `promote_staging(table)` (platform serving template step; the serving template is the platform's shared pipeline-to-website pattern from step 0.4, platform design doc).

C7 Lookup API (3A.4)

- Job: answer a point lookup from published Parquet with DuckDB in Lambda.
- Inputs: lookup request (see section 10); published Parquet.
- Outputs: JSON lookup answer; see openapi.yaml.
- Main functions and classes: `lookup_handler(event)` (FastAPI route on Lambda); `point_to_cell(lat, lon)` (resolution 8); `neighborhood_cells(cell, k)` (k is OQ-5); `query_cell_summary(cells)`; `nearest_buoy_latest(cell)`; `build_credits(sources)`.

C8 Lookup page (3A.4)

- Job: point entry, map, answer card with credits, status of cold start.
- Inputs: platform site template (map component, lookup form, answer card, status page); Lookup API.
- Outputs: the public page, with a preview deploy per pull request.
- Main components: `OceanLookupForm`, `OceanAnswerCard`, `CreditsFooter`, reused `MapView` (names for the reused ones are set by Project 1). Point entry method is OQ-9.

Phase B components

C9 AIS capture (3B.1)

- Job: a daily one-hour live AIS capture writing raw messages to S3 only (Kafka is off then).
- Inputs: AISStream key from Parameter Store; subscription (bounding boxes, optional filters; OQ-13); the daily start time (OQ-12).
- Outputs: raw capture files in S3 (format, part size: OQ-12), listed in a manifest with hash.
- Main functions and classes: `AisStreamClient` (shared websocket client); `run_capture(duration, sink)`; `RawCaptureWriter.write_part(messages)`; `build_subscription(bbox, filters)`.

C10 AIS live producer (3B.1)

- Job: during sessions, read the AISStream websocket and produce each message to Kafka.
- Inputs: AISStream key; subscription.
- Outputs: Kafka topic of raw messages.

- Main functions and classes: `AisStreamClient` (same shared class as C9); `KafkaSink.send(message)`; `run_live_producer(subscription)`. The connection comes from the session box, never a browser; connections used at once never exceed three (ADR 0008).

C11 AIS stream job (3B.1)

- Job: one Structured Streaming job used for both live sessions and the nightly replay (availableNow trigger).
- Inputs: Kafka topic (live or replayed from capture files); explicit schema; checkpoints on S3.
- Outputs: Iceberg `ais_messages` (table-ais-messages); lag and messages-per-second metrics.
- Main functions and classes: `ais_stream_job(mode)` with `mode` in live or replay; `AIS_MESSAGE_SCHEMA` (explicit schema, from OQ-13); `parse_ais_message(row)`; `dedupe_events(df)` (key and watermark: OQ-14, ADR 0007); `replay_capture_to_kafka(capture_manifest)`; `record_stream_metrics(batch)`.

C12 Buoy history backfill (3B.2)

- Job: backfill buoy history by year past the 45-day window; log missing stations and years; maintain SCD Type 2 station metadata.
- Inputs: NDBC historical files (source-ndbc-history); station metadata (source-ndbc-stations).
- Outputs: `buoy_readings` history, `buoy_stations` SCD Type 2 (table-buoy-stations), a gaps log.
- Main functions and classes: `ndbc_history_backfill_dag` (runs in work sessions only); `fetch_station_year(station_id, year)`; `log_missing(station_id, year)`; `scd2_merge_stations(current, incoming, as_of)`.

C13 Snowflake marts (3B.2)

- Job: run the dbt Snowflake target and the refresh step at the end of each pipeline run.
- Inputs: Iceberg tables via Glue.
- Outputs: marts passing tests on Snowflake. Which metrics are defined in MetricFlow for Project 3 is in the proposed step list (requirements.md section 8, the only place the candidate metrics are listed; awaiting John's approval; OQ-21).
- Main functions: `refresh_snowflake_iceberg(table)` (platform refresh step); dbt target `snowflake` of C4.

C14 Vessel identity (3B.3): detailed in section 7.1. Job: golden vessel record from AIS observations. Inputs: `ais_messages`, MMSI prefix reference, review labels. Outputs: `vessel_golden` (SCD Type 2), candidate pairs, review items.

C15 Review queue (3B.3, 3B.5)

- Job: hold low-confidence matches and agent proposals; take John's decisions; store labels.
- Inputs: review items from C14, proposals from C17.
- Outputs: decisions in DynamoDB (table-vessel-review-queue); labels in the lake after the nightly copy (table-vessel-labels).
- Main functions and classes: `ReviewQueue` (`put_item`, `list_open`, `get_item`, `record_decision`); `review_handler` Lambda routes (see section 10); `ReviewPage` and `ReviewItemCard` (React); `copy_queue_to_lake()` (nightly).

C16 AI matching (3B.4): detailed in section 7.1. Job: candidate pairs from embeddings, LLM decisions, trained classifier, three-version comparison.

C17 Data-quality agent (3B.5)

- Job: read failed checks and low-confidence matches; propose fixes; never write directly.
- Inputs: check results (platform metrics and gate results), low-confidence matches, read-only allowlisted tools (platform rule 13).
- Outputs: proposals in the review queue (types: merge two vessels, mark a buoy faulty, open an issue); a log of proposals with John's decisions; approval rate.

- Main functions and classes: `DqAgent.run(case)`; `list_failed_checks()` (read-only MCP tool); `list_low_confidence_matches()` (read-only MCP tool); `create_proposal(kind, payload, evidence)` (the only write tool); `ProposalLog.record(proposal, decision)`; `approval_rate(window)`. Where it runs, how approved proposals are applied, and the tool list are OQ-19.

C18 Buoy fault detection (3B.6)

- Job: flag stuck, spiking or silent buoys; record which rule fired.
- Inputs: `buoy_readings`.
- Outputs: `buoy_flags` (table-buoy-flags); shown by C7 and C8 with the readings.
- Main functions and classes: `detect_stuck(series)`; `detect_spike(series)`; `detect_silent(series, expected_interval)`; `BuoyAnomalyModel` (scikit-learn; algorithm and features: OQ-20); `flag_buoys(readings)`. No generated note (ADR 0014).

Phase C components (listed for completeness; designed in the second round)

- C19 MarineCadastre month loader: Spark job per month, measure input size, rows, runtime and cost.
- C20 Multi-node run: 3 to 4 node spot cluster for one month; compare to one node.
- C21 Global GEBCO: C2 extended to the global grid.
- C22 Vessel activity clusters and C23 batch briefings with number rejection.

7.1 Function-level detail for the trickiest part: vessel identity matching (3B.3 and 3B.4)

Module C14: vessel identity (3B.3). Data classes first, then functions.

Class	Purpose	Fields (design-time; extra fields set by OQ-13)
`Observation`	One AIS-derived record of a vessel at a time	mmsi, imo (nullable), ship_name, event_time_utc, latitude, longitude, message_type, source_key
`MmsiCheck`	Result of MMSI validation	valid_structure, country, prefix_known
`CollisionEvent`	One MMSI seen in two places too far apart	mmsi, obs_a, obs_b, implied_speed
`CandidatePair`	Two observation clusters that might be one vessel	pair_id, left_id, right_id, source (rules or embedding)
`RuleScore`	Rule evidence for a pair	score, reasons (list of named rules that fired)
`Route`	Where a pair goes	one of auto_match, review, no_match
`GoldenVessel`	One SCD Type 2 row of the golden table	vessel_key, mmsi_set, imo, names, valid_from, valid_to, is_current, evidence
`Label`	John's decision on a pair	pair_id, decision, decided_at, decided_by, source (queue or proposal)

Function or class	Purpose	Takes	Returns
`check_mmsi(mmsi, prefix_table)`	Validate MMSI structure and look up the country from its prefix	An MMSI, the prefix reference table (OQ-15)	`MmsiCheck`
`check_imo(imo)`	Validate an IMO number by its check digit	An IMO number	True or False
`implied_speed(obs_a, obs_b)`	Speed needed to travel between two positions in the elapsed time	Two observations	Speed value (unit fixed in code; limit in OQ-16)
`find_mmsi_collisions(observations, max_speed)`	Detect one MMSI used by two ships at once: consecutive positions with impossible speed	Observations for an MMSI sorted by event time, a maximum plausible speed (OQ-16)	List of `CollisionEvent`
`split_colliding_mmsi(observations, collisions)`	Separate the track of a colliding MMSI into two tracks so it is not merged as one vessel	Observations, collision events	Observations tagged with a track id
`normalize_name(raw_name)`	Standardize a ship name for comparison (rules fixed at build, OQ-16)	A name	A normalized name

Function or class	Purpose	Takes	Returns
`name_variant_score(name_a, name_b)`	Similarity of two normalized names with a named measure (OQ-16)	Two normalized names	Score from 0 to 1
`build_candidate_pairs_rules(observations)`	Propose pairs by shared MMSI, shared IMO or similar names (blocking keys fixed at build)	Observations	List of `CandidatePair`
`score_pair_rules(pair, observations)`	Combine MMSI, IMO, name and track evidence into a score and the list of rules that fired	A pair, observations	`RuleScore`
`route_pair(rule_score, thresholds)`	Decide auto match, review or no match (thresholds: OQ-16)	A score, thresholds	`Route`
`build_golden_record(cluster, labels)`	Merge a cluster of accepted observations into one record, choosing values by documented survivorship rules	Observations in a cluster, labels	`GoldenVessel`
`apply_labels_scd2(current_rows, labels, as_of)`	Apply John's decisions as SCD Type 2 rows: close old rows, open new rows	Current golden rows, labels, as-of time	New and closed `GoldenVessel` rows
`enqueue_for_review(pair, evidence)`	Put a low-confidence pair into the review queue with its evidence	A pair, evidence	Queue item id
`ReviewQueue.record_decision(item_id, decision)`	Store John's decision	Item id, decision	The stored `Label`
`evaluate_matches(predicted_pairs, labeled_pairs)`	Precision and recall on the hand-labeled pairs	Predicted pair decisions, 200 labeled pairs	Precision, recall
`duplicate_rate(vessel_records)`	Share of records that refer to a vessel already present (definition: OQ-17)	Vessel records before or after matching	A rate
`run_identity_resolution(version)`	Run one matching version end to end for comparison	Matcher version name	Golden rows, routed pairs, metrics

Module C16: AI-assisted matching (3B.4).

Function or class	Purpose	Takes	Returns
`EmbeddingIndex`	LanceDB index over vessel descriptions in S3 for candidate search	Vessel records	An index with `query(record, k)`
`embed_vessel_record(record)`	Turn a vessel record into an embedding (embedding model: OQ-16)	A vessel record	A vector
`find_candidates_by_embedding(index, record, k)`	Candidate matches by nearest neighbors	Index, record, k (OQ-16)	List of `CandidatePair`
`select_ambiguous_pairs(scored_pairs, band)`	Choose pairs the rules cannot decide (band: OQ-16)	Scored pairs, band	Pairs to send to the LLM
`build_pair_prompt(pair, prompt_version)`	Versioned prompt asking whether two records are one vessel, asking for a reason and a confidence	A pair, a prompt version in git	A prompt
`submit_match_batch(prompts)`	Send prompts through the model batch API	Prompts	Batch id
`collect_match_batch(batch_id)`	Read batch results and log cost	Batch id	List of `LlmDecision`
`LlmDecision`	Structured output	match (true or false), reason, confidence	n/a
`validate_llm_decision(raw)`	Reject malformed structured output	Raw model output	`LlmDecision` or an error

Function or class	Purpose	Takes	Returns
`build_training_set(llm_decisions, human_labels, excluded_pair_ids)`	Training rows from LLM decisions and John's labels, excluding the evaluation pairs (OQ-17)	Decisions, labels, excluded ids	Training rows
`pair_features(pair, observations)`	Numeric features of a pair (set at build)	A pair, observations	Feature vector
`train_pair_classifier(training_set)`	Train the small classifier with scikit-learn (trained on Databricks Free Edition with manual export of results and model, ADR 0012; algorithm set at build)	Training rows	A fitted model
`score_pair_classifier(model, pair)`	Score a pair without any model call	Model, pair	Match probability
`cost_per_1000_pairs(call_log, n_pairs)`	Cost per 1,000 pairs from the platform call log	Call log rows, pair count	A cost
`compare_versions(eval_pairs, results_by_version, call_log)`	One table with precision, recall, cost per 1,000 pairs and review-queue volume for rules only, rules plus AI, and the classifier	Evaluation pairs, per-version results, call log	Comparison table

The three versions (3B.4): version 1 uses C14 only; version 2 adds `find_candidates_by_embedding` and the batch LLM decisions for ambiguous pairs; version 3 replaces the LLM with `score_pair_classifier`, so the pipeline runs without model calls. The chosen version goes in a decision record (ADR 0012 is the placeholder until measured).

---

8. Runtime view

All times UTC. "Idempotent" means rerun-safe per platform rule 3.

8.1 Nightly run (Project 3 part)

The platform starts the session box at 10:00 UTC on spot, runs every pipeline and shuts it down when they finish, with a hard maximum of 2 hours. Proposed order of Project 3 tasks (the order is a design choice, not recorded):

```
sequenceDiagram
    participant AF as Airflow
    participant NDBC as NDBC
    participant GFW as GFW 4Wings
    participant S3 as S3
    participant SP as Spark
    participant KF as Kafka
    participant DBT as dbt
    participant GE as Great Expectations
    participant SN as Snowflake
    AF->>NDBC: fetch realtime files
    AF->>GFW: request report (one at a time)
    AF->>S3: write raw files with hash
    AF->>KF: replay the day's AIS captures
    SP->>KF: availableNow stream job, then stop
    SP->>S3: write ais_messages (Iceberg)
    AF->>SP: identity resolution, fault detection
    AF->>DBT: build staging, intermediate, marts
    DBT->>GE: gold tables
    GE-->>AF: pass or fail
    AF->>S3: publish Parquet only on pass
    AF->>SN: refresh Iceberg tables
    AF->>S3: copy review queue to lake
    AF->>AF: Iceberg maintenance (last task)
```

Failure handling:

- A non-pass from the gate stops publishing; the previous published data stays and the page shows it with a visible late notice and the last good run time (ADR 0016); an email goes out and an incident record is written (step 0.5).
- A Spark job over its row or runtime limit stops before publishing and opens an incident (hard limits table).
- Reaching the 2-hour maximum is recorded as an incident and the next run catches up.

- If spot is unavailable the night is skipped. The Fargate captures stay in S3 and are replayed by the next run (idempotent, QR-09).
- A GFW report over 100 seconds is recovered through last-report (ADR 0009). A failure after retries fails that task only; other tasks continue.
- A schema change in a source fails the contract diff before load (platform rule 5).

8.2 Daily AIS capture (always on)

```
sequenceDiagram
    participant EV as Schedule
    participant FG as Fargate task
    participant PS as Parameter Store
    participant AIS as AISStream
    participant S3 as S3
    EV->>FG: start (daily, time open: OQ-12)
    FG->>PS: read AISStream key
    FG->>AIS: subscribe with bounding box
    loop one hour
        AIS-->>FG: messages
        FG->>S3: write raw parts
    end
    FG->>S3: write manifest with hashes
```

Failure handling: a dropped websocket reconnects within the three-connection limit; a failed capture leaves a gap that is recorded and not backfilled, because AISStream is a live stream and missed hours cannot be fetched later. The task writes only to S3 (platform rule 16). A failed run emits the metrics row for the status page.

8.3 Live session stream

The producer (C10) reads AISStream and writes to Kafka; the stream job (C11) consumes and writes `ais_messages`. During a live session the job records lag (median and 95th percentile) and messages per second. Kafka runs only while the session box is on. If a live session overlaps the daily capture, the live producer and the capture task both read the same AISStream feed, using two of the three allowed connections (ADR 0008). The stream job removes the resulting duplicate messages with the same deduplication key (ADR 0007).

8.4 Lookup request (session box off)

```
sequenceDiagram
    participant V as Visitor
    participant CF as CloudFront and page
    participant API as API Gateway
    participant LM as Lambda (DuckDB)
    participant PQ as Published Parquet (S3)
    V->>CF: open page, enter a point
    CF->>API: GET lookup (lat, lon)
    API->>LM: invoke (cold start possible)
    LM->>PQ: read ocean_cell_summary, ocean_buoy_latest
    LM-->>API: JSON answer with credits and data times
    API-->>CF: response
    CF-->>V: answer card
```

Failure handling: a point outside the covered region or on land returns a clear message (400 or 404 in openapi.yaml); old published data is returned with a late notice and the last good run time, never hidden (ADR 0016); a missing Parquet or Lambda error returns 503 and the page shows an error state; the first request after inactivity may take up to 10 seconds (the stated page wording, step 0.4).

8.5 Identity resolution and review

Each nightly run builds candidate pairs (rules; plus embeddings in version 2), routes each pair (auto match, review, no match), writes review items to DynamoDB through C15, and applies the labels John has recorded since the last run as SCD Type 2 rows. John's decisions are stored as labels; the queue is copied to the lake as part of the nightly run. Failure handling: a failed run leaves the previous golden table current; the same labels applied twice produce the same SCD rows (idempotent: keyed on label id and as-of time).

8.6 Data-quality agent proposal

The agent (C17) reads failed checks and low-confidence matches through read-only tools. It calls `create_proposal`, which writes one item to the review queue, and nothing else. John approves or rejects on the review page. The decision and the approval rate are logged. Failure handling: a model error or malformed structured output creates no proposal and is logged; text found in source data (vessel names, check messages) is treated as data, never instructions (platform rule 13).

8.7 Buoy history backfill (work session only)

Backfills never run in the nightly run (decision "Backfills"). The DAG iterates stations and years, fetches NDBC history, loads each year as an overwrite of its partition (a slice of a table that can be rewritten on its own), and logs stations and years with no data. A rerun overwrites only the affected partitions.

---

9. Data model

Table names are design-time identifiers. Physical naming follows the platform's conventions (platform design doc, OQ-27). Every table has a contract in contracts/.

9.1 Layers and flow

Raw files (S3, immutable, hash and manifest) -> Iceberg lake tables (Glue database per environment) -> dbt staging, intermediate and marts -> published Parquet (lookup) and Snowflake marts (Phase B).

```
flowchart TB
    subgraph src["Sources"]
        direction LR
        s1["NDBC"]:::ext
        s2["GEBCO 2026"]:::ext
        s3["GFW 4Wings"]:::ext
        s4["AISStream"]:::ext
        s5["`MMSI prefix
reference`"]:::ext
    end
    raw[("Raw files: S3, immutable, hash and manifest")]:::store
    subgraph lakeg["Iceberg lake tables (Glue)"]
        direction LR
        t1["`buoy_readings
buoy_stations
buoy_flags`"]:::cont
        t2["seafloor_cells"]:::cont
        t3["fishing_cells"]:::cont
        t4["ais_messages"]:::cont
        t5["`vessel_golden
vessel_pairs
vessel_labels`"]:::cont
    end
    rq[("`vessel_review_queue
DynamoDB`")]:::store
    subgraph bld["Build and publish"]
        direction LR
        dbt["`dbt: staging,
intermediate, marts`"]:::sys
        gate["`Great Expectations
publish gate`"]:::sys
    end
    subgraph pub["Published"]
        direction LR
        p1[("`ocean_cell_summary
ocean_buoy_latest
Parquet`")]:::store
        p2[("`Snowflake marts
Phase B`")]:::store
    end
    s1 --> raw
    s2 --> raw
    s3 --> raw
    s4 --> raw
    s5 --> raw
```

```
    raw --> t1
    raw --> t2
    raw --> t3
    raw --> t4
    t4 --> t5
    t5 -->|"low-confidence pairs"| rq
    rq -->|"nightly copy of decisions"| t5
    t1 --> dbt
    t2 --> dbt
    t3 --> dbt
    t5 --> dbt
    dbt --> gate
    gate -->|"pass, read by the Lookup API"| p1
    t1 -.->|"read in place"| p2
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef cont fill:#d1fae5,stroke:#047857,color:#052e16
    classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
    classDef sys fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

9.2 Sources

Source	Contract	Used by
NDBC realtime	source-ndbc-realtime	C1
NDBC history	source-ndbc-history	C12
NDBC station metadata	source-ndbc-stations	C1, C12
GEBCO 2026 grid and source-type grid	source-gebco-grid	C2
Global Fishing Watch 4Wings	source-gfw-4wings	C3
AISStream messages (live and capture)	source-aisstream	C9, C10, C11
MMSI country prefix reference (not in the plan's source list; OQ-15)	source-mmsi-prefix-reference	C14

9.3 Tables

Table	Kind	Grain / key	Contract
`buoy_readings`	Iceberg	station, event time	table-buoy-readings
`buoy_stations`	Iceberg, SCD Type 2	station, valid_from	table-buoy-stations
`seafloor_cells`	Iceberg	H3 res 8 cell	table-seafloor-cells
`fishing_cells`	Iceberg	cell, date, layer (AIS fishing or SAR unmatched), one partition per pull	table-fishing-cells
`ais_messages`	Iceberg	source key	table-ais-messages
`vessel_golden` and link table	Iceberg, SCD Type 2	vessel_key, valid_from	table-vessel-golden
`vessel_pairs`	Iceberg	pair id, matcher version	table-vessel-pairs
`vessel_labels`	Iceberg (nightly copy of decisions)	label id	table-vessel-labels
`vessel_review_queue`	DynamoDB, point-in-time recovery on	item id	table-vessel-review-queue
`buoy_flags`	Iceberg	station, flag time, rule	table-buoy-flags
`ocean_cell_summary`	Published Parquet (gold)	H3 cell	table-ocean-cell-summary
`ocean_buoy_latest`	Published Parquet (gold)	station	table-ocean-buoy-latest

Entity-relationship view of the published tables and the tables they are built from (key columns only; full columns are in the contracts). Joins between cells use `h3_cell`; fishing cells may be at a different resolution and join through `cell_to_parent` (section 9.4).

```
erDiagram
    seafloor_cells {
        string h3_cell PK
        boolean is_mapped
        double mapped_pixel_share
    }
    fishing_cells {
        string h3_cell PK
        date report_date PK
        string layer PK
        double fishing_hours
        double detections
    }
    buoy_stations {
        string station_id PK
        timestamp valid_from_utc PK
        string h3_cell
    }
    buoy_readings {
        string station_id PK
        timestamp observation_time_utc PK
        double wave_height
        double water_temperature
    }
    buoy_flags {
        string station_id PK
        timestamp flagged_at_utc PK
        string flag_type
        string rule_id
    }
    ocean_cell_summary {
        string h3_cell PK
        boolean is_mapped
        double ais_fishing_hours
        double possible_dark_vessel_detections
        string nearest_station_id FK
    }
    ocean_buoy_latest {
        string station_id PK
        timestamp observation_time_utc
        string flag_type
    }
    seafloor_cells ||--o| ocean_cell_summary : "seafloor flag per cell"
    fishing_cells }o--o| ocean_cell_summary : "counts per cell"
    buoy_stations ||--o{ buoy_readings : "station_id"
    buoy_stations ||--o{ buoy_flags : "station_id"
    buoy_readings }o--|| ocean_buoy_latest : "latest per station"
    buoy_flags }o--o| ocean_buoy_latest : "flag per station"
    ocean_buoy_latest ||--o{ ocean_cell_summary : "nearest station"
```

Vessel identity tables (Phase B):

```
erDiagram
    ais_messages {
        string source_key PK
        string mmsi
        timestamp event_time_utc
    }
    vessel_identity_link {
        string mmsi PK
        string vessel_key FK
        timestamp valid_from_utc PK
    }
    vessel_golden {
        string vessel_key PK
        timestamp valid_from_utc PK
        string imo
        string matcher_version
    }
    vessel_pairs {
        string pair_id PK
        string matcher_version PK
        string route
    }
    vessel_review_queue {
```

```
        string item_id PK
        string kind
        string status
    }
    vessel_labels {
        string label_id PK
        string item_id FK
        string pair_id FK
        string decision
    }
    ais_messages }o--o| vessel_identity_link : "mmsi"
    vessel_identity_link }o--|| vessel_golden : "vessel_key"
    vessel_pairs ||--o{ vessel_review_queue : "low-confidence pairs"
    vessel_review_queue ||--o{ vessel_labels : "decisions copied nightly"
    vessel_pairs ||--o{ vessel_labels : "pair_id"
```

Platform tables used but defined in platform contracts: the run metrics table (`platform_run_metrics`, step 0.5). The call log table (`llm_calls`) is not in the platform contracts yet; it needs a contract there before Project 3 Phase B.

9.4 Data rules

- UTC for every timestamp (platform rule 15).
- Event time and source key on every record (platform rule 2).
- H3: seafloor cells at resolution 8 (ADR 0002); fishing cells at the resolution chosen from the GFW grid (OQ-6); cross-source joins with `cell_to_parent`.
- History in rows (SCD Type 2) for station metadata and golden vessels (ADR 0010, ADR 0011).
- Nothing in Project 3 stores visitor identifiers.

---

10. API

One public endpoint (lookup) and the owner-only review routes. Full specification: openapi.yaml. Route names are provisional (OQ-10).

Endpoint (provisional)	Method	Purpose	Gateway type	Notes
`/lookup`	GET	Answer for a point: unmapped share, fishing activity, nearest buoy's latest conditions (with flags in Phase B), credits and data times	HTTP type (public pages)	No key; throttling limits open (OQ-10)
`/review/items`	GET	List open review items	REST type, one API key (ADR 0015)	Owner only
`/review/items/{item_id}`	GET	One item with evidence	REST type, one API key (ADR 0015)	Owner only
`/review/items/{item_id}/decision`	POST	Record John's decision	REST type, one API key (ADR 0015)	Owner only; the only write through the API

The data-quality agent does not use these routes. Its single write is `create_proposal` (section 7, C17), which puts one item in the queue with a role limited to that action.

---

11. Deployment view

Part	Runs on	Environment notes
Lookup page	CloudFront and S3	Preview deploy for every pull request; dev and live separate
Lookup API and owner-only review routes	Lambda behind API Gateway	Reserved concurrency and throttling: OQ-10
Published Parquet	S3	Written only after the gate passes

Part	Runs on	Environment notes
Review queue	DynamoDB, point-in-time recovery on	Dev and live separate
AIS capture	Fargate scheduled task	About $1 a month for all Fargate ingest (estimate); writes only to S3
Airflow DAGs, Spark jobs, dbt, GE, data-quality agent (if run in pipeline), embeddings	Session session box on k3s (t4g.2xlarge or m7g.2xlarge)	Started by Terraform per session and nightly; 4-hour session limit; 2-hour nightly maximum
Kafka	Session box, one broker	Only while the session box is on
Marts	Snowflake (3-credit monthly resource monitor, time limit on every query)	Phase B
Keys	Parameter Store	GFW, AISStream, model API
Phase C multi-node run	3 to 4 node spot cluster for one month of data	Second round

Schedules.

Schedule	Time	Source
Nightly run	10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern)	Default used, Build Plan Decisions
Daily AIS capture	One hour, start time open	OQ-12
Weekly smoke test	Weekly, pulls one small sample from every source	Step 1B.9 (shared)
Backfills	Work sessions only	Decision "Backfills"

Environments. Dev and live have fully separate buckets, Glue databases, DynamoDB tables and roles; dev roles cannot write live data. Only merges to main reach live, after the changed pipelines pass on sample data in CI. Dev raw files are not Object Locked (so the 14-day clean-up works); live raw files are locked 30 days.

---

12. Cross-cutting concerns

12.1 Security

- Keys (Global Fishing Watch, AISStream, model API) only in Parameter Store (SecureString), read by role. No keys in code or GitHub.
- One least-privilege IAM role per pipeline. The capture task role may read its key and write only to the capture prefix. The agent role may read and call `create_proposal` only (ADR 0013).
- The AISStream key never reaches a browser (ADR 0008).
- gitleaks and Trivy scans run in CI (step 0.1).
- Source text (vessel names, messages) is data, never instructions (platform rule 13).
- Owner-only review routes: owner only, behind one API key held in Parameter Store, keyed REST type; the review page is not linked from the public site (ADR 0015, decided by John, Oct 9, 2026).

12.2 Privacy

The project collects no visitor data. The review queue holds vessel data and John's decisions. Vessel names and MMSI stay internal and are never shown publicly (ADR 0008). The data inventory and deletion periods belong to the Security and privacy document (written while building).

12.3 Observability

Freshness and row counts per run to the metrics table, a status page, an email on missed freshness or failure and one incident record each (step 0.5). Stream lag (median, 95th percentile) and messages per second per live session. Model calls logged with model, prompt version, tokens, latency and cost. Traces to X-Ray for agent runs (platform AI foundations).

12.4 Cost limits

Item	Hard limit	Status
Each Spark job	Maximum rows read and written and maximum runtime; over either, stop before publishing and open an incident	Numbers set from measured sizes (OQ-1, OQ-23)
Servers	4-hour session limit, idle shutdown, 2-hour nightly maximum	Decided
Snowflake	3 credits a month; time limit on every query	Decided
AI model	$15 monthly; daily cap; token limit per answer	Decided; shared with other projects
S3	Clean-up rules (dev 14 days, temporary 7 days, failed uploads 7 days, pipeline logs 90 days, security logs 1 year, old versions 30 days) and a size alarm	Decided (shared step 1B.9)
DynamoDB	Provisioned within the free tier	Decided
Everything on AWS	$40 budget as backup	Decided
Phase C cluster	"A few dollars" for one month on 3 to 4 nodes	Build plan, Phase C wording only; the dollar limit is open (OQ-23) and the cost is measured later

12.5 Reliability

Idempotent, atomic tasks; one partition per pull; replay-safe stream (ADR 0006, ADR 0007); publish only after gate passes; the lookup does not depend on the session box. DynamoDB point-in-time recovery lets a bad write or deletion be undone.

12.6 Licensing and credit

Global Fishing Watch: non-commercial, credit on page (ADR 0009). AISStream: derived maps and counts with credit (ADR 0008). NDBC, GEBCO and MarineCadastre terms and credit wording: OQ-24 (MarineCadastre is CC0). Licensing must be checked before charging for the project.

---

13. Testing strategy

Tests use small sample files in CI, like ordinary unit tests; there are no staged production failures (build plan "Testing").

Level	What	Examples
Unit	Pure functions	`classify_source_code` on codes 0, 10 to 17, 40 to 46, 70 to 72; `point_to_cell`; `check_imo`; `check_mmsi`; `implied_speed`; `find_mmsi_collisions` on a sample track with a known collision; `scd2_merge_stations`; `detect_stuck`, `detect_spike`, `detect_silent`
Pipeline (Spark on sample data in GitHub Actions)	Jobs end to end on samples	GEBCO sample tile to `seafloor_cells`; sample GFW response to `fishing_cells`; sample AIS capture replayed twice (AC-3B.1-3); job over its row limit stops before publish
Data tests	dbt tests, Great Expectations	`unique` and `not_null` on keys; range checks on waves and temperature; freshness; empty result; gate failure stops publish
Contract tests	Schema diff	Each source sample validated against its contract; weekly smoke test on live sources
Evals	Matching and agent	Precision and recall on the 200 hand-labeled pairs for all three versions; red-team set (prompts written to make the agent misbehave) including instructions hidden in source text for the agent; thresholds set from the proposed step list (requirements.md section 8, awaiting approval)
Security tests	Permissions	Dev role refused when writing live data; agent role refused for any write other than `create_proposal`; gitleaks clean
End-to-end smoke	Lookup	Request to the dev deploy returns the expected shape; preview link per pull request

Level	What	Examples
First-run checks	Assumptions before building	Phase A: Global Fishing Watch key works; GEBCO source-type codes read correctly for the region. Phase B: first AISStream connection runs and its message rate is recorded (three connections; not from a browser). Each check has a fallback if it fails; the fallbacks are not defined in the plan (OQ-13)
Reference docs	Docstrings	Every function carries a docstring; pdoc or Sphinx published

14. Alternatives considered

Topic	Options weighed	ADR
Phase A region	Gulf of Mexico (example in plan), others	0001
Seafloor grid resolution	H3 res 8 (chosen); res 9, res 4, native raster	0002
Meaning of "mapped"	GEBCO codes 10 to 17 only (chosen); include 40 to 46	0003
Dark vessel wording	"possible dark vessels" (chosen); "dark vessels"	0004
Map and serving	Reuse Project 1 and the serving template (chosen); new map	0005
Live stream design	Kafka live with nightly replay and Fargate capture (chosen); always-on server; Fargate to Kafka	0006
Event identity and dedup	Source key with dropDuplicatesWithinWatermark (proposed)	0007
AISStream usage	Server side only, within three connections, derived outputs	0008
GFW handling	One request at a time; non-commercial credit	0009
Buoy history	Rows with SCD Type 2 and yearly backfill; snapshots	0010
Identity resolution	Golden record with review queue, labels as SCD Type 2	0011
Matching method	Rules only; rules plus AI; trained classifier (comparison pending)	0012
Agent permissions	Proposals only (chosen); direct writes	0013
Buoy faults	Rules plus scikit-learn, no generated note (chosen); LLM notes	0014
Review queue access	Owner only behind one API key, keyed REST type (chosen); open to the public site; separate login service	0015
Late or failed runs	Show old data with late notice and last good run time (chosen); hide expired data	0016

Platform decisions that Project 3 relies on are linked, not repeated: platform ADRs (k3s over Docker Compose; DuckDB serving; Snowflake over Iceberg; DynamoDB for visitor data; Parameter Store; Terraform; Flink, Trino, BigQuery, pgvector and DataHub removed).

Minor decisions folded from ADRs

These were separate ADRs and are recorded here in full because each is a setting, a wording rule, a format or a repeat of a platform rule. The old numbers are kept so references still resolve.

Old ADR	Decision	Detail
ADR-0002	Seafloor grid on H3 resolution 8	Convert the region cut of the GEBCO grid to H3 resolution 8 and store one row per ocean cell. Join to other sources (fishing activity, buoy stations) at the coarser of the two resolutions through `cell_to_parent`.
ADR-0004	Label SAR detections without an AIS match "possible dark vessels"	Use the label "possible dark vessels" for SAR detections with no AIS match, on the page, in the README and in table and column descriptions. The word "possible" is never dropped. The label wording appears on the page (3A.3 Done when).
ADR-0005	Reuse Project 1's map and the platform serving template	The Project 3 lookup uses the platform serving template and Project 1's map component. Gold tables are published as Parquet; the lookup is a Lambda function with DuckDB behind the HTTP type of API Gateway (public pages). The cold-start sentence from step 0.4 appears on the page.

Old ADR	Decision	Detail
ADR-0010	Buoy history kept in rows, backfilled by year in work sessions	History is stored as rows in `buoy_readings`, partitioned so that a rerun of a station-year overwrites only that partition. Station metadata is stored as SCD Type 2 rows (`buoy_stations`). The backfill DAG is triggered by John during a work session, never scheduled into the nightly run. Every missing station or year is written to a gaps log, not silently skipped.
ADR-0016	Late or failed nightly run: show old data with a late notice	When the nightly run is late or failed, the page shows the old data with a visible late notice and the time of the last good run. Data is never hidden. Every answer keeps its "as of" time for the buoy readings and the fishing data. How "late" is determined and the notice wording are open (OQ-29).
ADR-0017	MMSI country prefix reference stays open until a published table is chosen	The prefix reference stays an open question. No prefix list is typed in, copied from memory or guessed (John, Oct 9, 2026). The source is chosen only when it is a published reference table, and its licence is read from the publisher's own page before any of its data is used (platform rule on sources; OQ-24 covers the wording). Until then, `check_mmsi` runs with a labelled test table in tests only, and the lookup and review pages show "country unknown" rather than a guessed country.
ADR-0019	Phase C done-when checks are recorded as missing until Phase C is designed	The Phase C done-when checks are recorded as missing. They are not invented in this document set. They are written when Phase C is designed, after Phase B has reached its checks. Phase C is not marked done, and its requirements are listed as incomplete in requirements.md.
ADR-0020	The 200 evaluation pairs are never used for training or as prompt examples	The 200 evaluation pairs are never used to train the classifier. They are never used as examples in LLM prompts. Other labels John makes may train the classifier, because they are not in the evaluation set.
ADR-0021	Say "latest buoy reading" with its time, not "current conditions"	The page and the documents say "latest buoy reading", with the time it was taken. "Current conditions" is not used for buoy data. The word "current" stays where it means a database row (`is_current`).
ADR-0022	IP addresses are removed from request logs before they are kept	Request logs may be written, because they are needed for abuse limits and for finding faults. Addresses are removed (or truncated to the network part only) before anything is stored beyond the short delete window used by Project 1. No address is written to the warehouse, the lake, the call log or any metrics table. The Terraform for the logs sets the removal step; a test checks that a stored log sample has no address field.
ADR-0023	Licence and credit wording is checked on the provider's own page	Every licence or credit statement in the register cites the provider's own page, with the date it was read. A statement found only on a third-party page is marked "secondary only". It is not used as the credit wording. A statement that cannot be found or read is marked "unverified". Nothing is guessed. Where the page is blocked, the register says so. Where John must decide before anything is published from an unverified source, the item goes to the "Decisions for John" section of security-and-privacy.md.

15. Risks

Risk	Impact	Handling
AISStream message rate is unknown	Jobs, storage and cost limits cannot be set	OQ-1: first-run check records the rate; limits set after
AISStream has no terms page	Allowed use is unclear	Maps and counts only, vessel names and MMSI internal; John will ask AISStream about their terms and the rule is loosened only after their answer (ADR 0008; OQ-3)
AISStream connection limit of three	Live producer, capture and smoke test can collide	Count connections; capture and live session use at most two; smoke test uses a brief connection (ADR 0008)
Live session and capture produce duplicates of the same stream	Double-counted vessels	Dedup by source key (ADR 0007)
GFW is non-commercial and limits to one report at a time	Licensing; run stalls	Credit and non-commercial note; serialized requests with last-report recovery (ADR 0009)
"Possible dark vessels" read as confirmed	Misleading public claim	Fixed wording on the page (ADR 0004)

Risk	Impact	Handling
Buoy "latest reading" is up to a day old (ADR 0021)	Misleading freshness	Show data time, and a visible late notice with the last good run time when late, never hide the data (ADR 0016); OQ-7, OQ-29
Mapped rule hides uncertainty (interpolated cells not counted as mapped)	Misread seafloor knowledge	Rule written down, codes stated on the page (ADR 0003)
MMSI prefix reference source and IMO field are unconfirmed	Validation may be impossible as specified	OQ-13, OQ-15
Evaluation pairs leak into training	Inflated metrics	Hold out (proposed, OQ-17)
LLM spend exceeds plan	Surprise cost	$15 monthly limit, batch calls, cheapest model, call log (QR-03)
Agent acts on injected text	Wrong change	Proposals only, read-only tools, source text as data (ADR 0013)
Review queue exposed publicly	Unauthorized decisions	Owner only behind one API key; page not linked from public site (ADR 0015)
Spot unavailable	Night skipped	Captures wait in S3; next run replays; idempotent
Nightly maximum reached	Incomplete run	Recorded as incident; next run catches up
Snowflake refresh does not show new Iceberg data	Marts stale	First-run check in shared step 1B.3; shared with Project 1
Raw bucket growth from AIS captures	S3 cost	Measured size; storage alarm; Object Lock on live raw only

16. Open questions

Each question needs a spike or a decision and has a matching GitHub issue (placeholder "issue: to be created"). Answered questions stay in this table, with their answer and ADR number added, so existing references still resolve.

ID	Question	What answering it needs	GitHub issue	Answer and ADR (Oct 9, 2026)
OQ-1	What is AISStream's message rate for the region? (From the Documentation Plan's list of questions still open.)	John's AISStream key and a live connection	issue: to be created	Open.
OQ-2	Which region and bounding box is Phase A? The plan says "for example the Gulf of Mexico".	John's choice (ADR 0001)	issue: to be created	Open. The region is still John's choice (ADR 0001 is proposed).
OQ-3	What does AISStream say about its terms? John will ask them. Until they answer, only maps and counts are shown and vessel names and MMSI stay internal; the rule is loosened only after their answer.	John's question to AISStream and their reply	issue: to be created	Partly answered: display rule decided (ADR 0008; maps and counts only). The AISStream terms reply is still open.
OQ-4	How do pixel source-type codes combine into one H3 res 8 cell flag, and how is "unmapped share" computed (share of unmapped pixels? any unmapped pixel?)	Prototype on the region cut; John's choice; write the rule in 3A.2	issue: to be created	Open.
OQ-5	What is "nearby" for the unmapped share and fishing activity (ring of cells, radius, or fixed grid parent)?	John's choice	issue: to be created	Open.
OQ-6	What are the Global Fishing Watch dataset(s), temporal resolution, grid resolution, data delay and the exact credit wording, and which H3 resolution stores the counts (rule 9 picks the cell just larger than the grid)?	First test report with John's key	issue: to be created	Open.

ID	Question	What answering it needs	GitHub issue	Answer and ADR (Oct 9, 2026)
OQ-7	NDBC: which product lists the region's stations; field names; which measurements besides waves and temperature appear on the page; range-check bounds for waves and temperature; how often the realtime files are refreshed relative to the nightly run	Download of sample station files	issue: to be created	Open.
OQ-8	Numeric freshness target for each table (contracts list cadence only)	John's decision; belongs to Operations docs	issue: to be created	Open.
OQ-9	How does a visitor enter a point (latitude and longitude, click on the map, place search)? Is a geocoder needed?	John's decision; Project 1 map component capabilities	issue: to be created	Open.
OQ-10	Route names, throttling and reserved-concurrency limits for the public lookup	Serving template (platform docs)	issue: to be created	Open.
OQ-11	Who can write review-queue decisions, and how is access to the review queue controlled?	John's decision	issue: to be created	Answered: owner only, behind one API key held in Parameter Store; page not linked from the public site (ADR 0015).
OQ-12	Daily capture start time, raw capture file format and part size	John's decision; size from OQ-1	issue: to be created	Open.
OQ-13	Which AISStream message types and fields are needed (position, static data), which carry IMO number and other static fields, subscription bounding boxes and filters; fallbacks for first-run checks	First live connection (same spike as OQ-1)	issue: to be created	Open.
OQ-14	Dedup key composition, watermark length, and the definition of "lag" for the 3B.1 measurement	Spike on live and replayed capture	issue: to be created	Open.
OQ-15	Source and license of the MMSI country prefix reference table (not in the plan's source list)	John's decision	issue: to be created	Open by John's decision: no prefix list is guessed (ADR 0017).
OQ-16	Matching parameters: maximum plausible speed; name normalization rules; name similarity measure; blocking keys; route thresholds; embedding model; embedding neighbors count; ambiguous band for the LLM; which concrete cheapest model	Experiments on sample data; platform model configuration	issue: to be created	Open.
OQ-17	Evaluation design: how the 200 pairs are sampled and labeled; held-out from training and prompts; definition of "duplicate rate"; pass thresholds for precision and recall	John's decision	issue: to be created	Partly answered: the 200 pairs are held out from training and prompts (ADR 0020, default; approval under OQ-30). Sampling, the duplicate-rate definition and thresholds are open.
OQ-18	Can the export of results and model from Databricks Free Edition be automated? (Manual export is decided.) Which algorithm and features does the classifier use?	A test on John's Databricks account; experiments	issue: to be created	Partly answered: manual export from Databricks Free Edition decided (ADR 0012). Whether export can be automated, and the algorithm and features, are open.
OQ-19	Data-quality agent: where it runs, trigger, tool list, and how an approved proposal is applied (merge, mark buoy faulty, open issue)	Design decision with John	issue: to be created	Open.
OQ-20	Buoy fault rules: thresholds for stuck, spiking and silent; scikit-learn model and features	Experiments on buoy history	issue: to be created	Open.

ID	Question	What answering it needs	GitHub issue	Answer and ADR (Oct 9, 2026)
OQ-21	Approval of the proposed step list for metrics, evals, tracing and MCP (requirements.md section 8), and the metrics, eval set sizes and thresholds it leaves to be set (the candidate metrics are listed only in that step list)	John's approval	issue: to be created	Partly answered: proposed step list written (requirements.md section 8; ADR 0018, proposed). Approval, metric set, sizes and thresholds are open.
OQ-23	What are the hard row and runtime limits per Spark job, and what dollar limit applies to the Phase C cluster run (the build plan says only "a few dollars")?	Measured sizes (OQ-1 and sample runs)	issue: to be created	Open.
OQ-24	Terms and credit wording for NDBC, GEBCO 2026 and Global Fishing Watch; license for AISStream	Reading each provider's terms	issue: to be created	Partly answered: the Global Fishing Watch and GEBCO terms were read on their own pages (ADR 0023; see security-and-privacy.md). AISStream, NDBC, MarineCadastre's own licence text and GEBCO's credit wording are not verified.
OQ-25	Data sizes of the GEBCO regional cut, a GFW report, buoy history and an AIS hour ("Measure first" in the plan)	Downloads	issue: to be created	Open.
OQ-26	Done-when checks for Phase C	Written when Phase C is designed	issue: to be created	Answered: recorded as missing until Phase C is designed (ADR 0019).
OQ-27	ODCS and OpenAPI versions, naming conventions and the contract template come from the platform documents	Platform docs	issue: to be created	Partly answered: the platform set uses ODCS v3.0.2 for contracts (platform changelog and contracts). The naming convention and contract template are not yet set in the platform documents.
OQ-28	Time zone shown for "as of" times on an ocean point page (the viewer's browser or something else)	John's decision	issue: to be created	Open.
OQ-29	Late-data display: how is "late" determined (the threshold after the expected nightly publish) and what is the exact notice wording? (The rule itself is decided: show old data with a visible late notice and the last good run time, never hide it.)	John's decision	issue: to be created	Partly answered: rule decided (old data with a late notice and the last good run time; ADR 0016). The late threshold and notice wording are open.
OQ-30	Approval of two proposed requirements that are not John decisions: FR-20 (an explicit "as of" time on every lookup answer) and FR-57 (the 200 evaluation pairs are never used for training or prompts). Also the proposed step list (OQ-21).	John's approval	issue: to be created	Open: awaiting John's approval of FR-20, FR-57 (ADR 0020, default) and the step list (ADR 0018, proposed).
OQ-31	How does the owner's API key reach the review page without being visible in browser code? A key placed in code the browser downloads can be read by anyone who opens the page. One option to decide on is that the owner types the key into the page at the start of each session and the page never stores it in code.	John's decision, before the review page is built	issue: to be created	Open. ADR 0015 decides who may write, not how the key reaches the page.

Counts at Oct 9, 2026: 30 questions in this table. 2 answered (OQ-11, OQ-26), 7 partly answered (OQ-3, OQ-17, OQ-18, OQ-21, OQ-24, OQ-27, OQ-29), 21 open. Answered questions keep their rows so that references still resolve.

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17. Glossary

Term	Meaning
AIS	Automatic Identification System; ships broadcast identity and position

Term	Meaning
AISStream	A free websocket service that streams AIS messages
availableNow	Spark trigger that processes everything new since the last run, then stops
Buoy / NDBC	NOAA National Data Buoy Center; stations report waves, temperature and more
C4	Context, container, component, code model for architecture diagrams
Dark vessel	A vessel not appearing in AIS. In this project only "possible dark vessels": SAR detections with no AIS match
Golden vessel record	The single cleaned record for one real vessel built from many observations
GEBCO	General Bathymetric Chart of the Oceans; the seafloor grid
GFW	Global Fishing Watch
H3	Hexagonal grid; resolution 8 is used for the seafloor grid
IMO number	International Maritime Organization vessel number, which carries a check digit
Iceberg	Table format on S3 used for the lake
LanceDB	Vector search files on S3
MMSI	Maritime Mobile Service Identity; the number a ship broadcasts; its first digits indicate a country
ODCS	Open Data Contract Standard
4Wings	Global Fishing Watch's gridded report API
SAR	Synthetic aperture radar; satellite detections of vessels
SCD Type 2	Slowly changing dimension: history kept as rows with valid-from and valid-to
Source-type identifier (SID) grid	GEBCO's grid saying what kind of data supports each cell
Session box	The spot EC2 session box on k3s started per session and nightly
Structured Streaming	Spark's streaming engine
Write, audit, publish	New data lands in staging, passes checks, then goes live
Glue	AWS Glue Data Catalog: the catalog that lists the Iceberg lake tables so every engine reads the same tables
CloudFront	AWS service that delivers the website files
API Gateway	AWS service that receives web requests and passes them to Lambda; the HTTP type serves public pages and the REST type supports API keys
Terraform	The tool that creates the AWS resources from code
Partition	A slice of a table that can be rewritten on its own
Manifest	A list of files with their URL, size and hash
Red-team	Prompts written to make an agent misbehave, used as a test set
Serving template	The platform's shared pattern from step 0.4: publish gold tables as Parquet, answer with DuckDB in Lambda behind API Gateway, show on a React site
Marquez	The platform's lineage graph tool; not used directly by Project 3 components
Airflow DAG	A scheduled pipeline definition in Airflow (DAG means directed acyclic graph: tasks with a fixed order and no loops)
Apache Spark	The engine that runs the batch and streaming jobs
As-of time	The time at which the data shown was taken or last published
Classifier	A trained scikit-learn model that takes features of two vessel records and says whether they are the same vessel, with no model API call
Cold start	The delay of the first request after a period of inactivity, while the service starts
dbt	Tool that builds tables from SQL models; layers are staging (cleaned copy of a source), intermediate (joins and calculations) and marts (final tables for use)

Term	Meaning
DuckDB	A database library that queries Parquet files; it runs inside the Lambda function
DynamoDB	AWS database that is always on; holds the review queue
Embedding	A list of numbers that represents a vessel record so similar records can be found by distance
Fargate	AWS service that runs a container on a schedule without a server to manage; runs the daily AIS capture
Gold table	A final, checked table that is published for the website
Great Expectations	Tool that runs data checks; used as the publish gate
Idempotent	Safe to run again: a second run gives the same result as the first
Kafka	Message queue that holds the AIS messages between the producer and the stream job; on only while the session box is on
Label	Two meanings. (1) A recorded correct answer about a vessel pair, made by John, used to apply changes and to train and check matching. (2) The fixed wording shown on a page, such as "possible dark vessels"
Lambda	AWS service that runs code on request without a server to manage; runs the lookup
Late notice	The visible message on the page, with the last good run time, when the nightly run is late or failed (ADR 0016)
LLM	Large language model; here only the cheapest model, called through the batch API
MCP	Model Context Protocol; the way tools are offered to an AI agent
MetricFlow	Tool in which metrics are defined once and computed on Snowflake and DuckDB
Nightly run	The scheduled run of every pipeline, started at 10:00 UTC on the session box
Object Lock	S3 setting that stops a file being deleted or overwritten for a set time (30 days on live raw files)
Parameter Store	AWS place where every key is kept, encrypted
Parquet	A column-based file format used for the published tables
Point-in-time recovery	DynamoDB setting that lets a bad write or deletion be undone
Precision and recall	Of the pairs the matcher called the same vessel, the share that really are (precision); of the pairs that really are the same vessel, the share the matcher found (recall)
Publish gate	The data checks that must pass before new data is published
Smoke test	A test that pulls a sample from a source and checks it still has the expected format; the weekly one does this for every source and alerts on a change
Snowflake	The cloud data warehouse that holds the marts
Spot	AWS spare capacity at a lower price that can be taken back with two minutes of warning
Unmapped share	The share of nearby seafloor not directly measured, by the rule in ADR 0003
Watermark	In streaming, the time limit after which late events are no longer waited for
X-Ray	AWS service that records traces of agent runs

Part 4: Operations

Ocean Map/operations.md

Project 3: Ocean map. Operations

Field	Value
Status	Draft for John's review (not yet approved). Numbers marked "open" are not set.
Owner	John
Last updated	Oct 9, 2026
Related	design-doc.md (sections 11 and 12), requirements.md (QR-04, QR-08), security-and-privacy.md, platform operations.md

The platform runs the session box, the nightly schedule, the status page and the shared runbooks. This document covers what runs for Project 3 and what to check. The platform's runbooks (PRB-1 to PRB-7) apply to Project 3 as well.

1. Schedules

Schedule	When	Notes
Nightly run	10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern)	Spot only. Two-hour hard maximum. A late or failed run is caught up by the next run.
Daily AIS capture	One hour a day; start time open (OQ-12)	Fargate task. Writes raw messages to S3 only. Kafka is off then.
Live stream	During work sessions only	Kafka and Structured Streaming on the session box, four-hour session limit.
Weekly smoke test	Once a week	Pulls one small sample from every source (platform shared step). Counts toward the AISStream connection limit.
Backfills	Work sessions only	Buoy history by year (ADR 0010).
Phase C monthly run	Second round	Spot cluster of 3 to 4 nodes; design written when Phase C starts (ADR 0019).

2. Freshness and what the page shows

- Each table has a freshness target. The contracts list the cadence; the numeric targets are open (OQ-8).
- A late or failed nightly run does not hide data. The page shows the old data with a visible late notice and the time of the last good run (ADR 0016). How "late" is decided is open (OQ-29).
- Buoy readings are the latest daily values, up to about a day old. The page says "latest buoy reading" with the time it was taken (ADR 0021).

3. Monitoring

What	Where	Alert
Freshness and row count for each DAG run	The platform metrics table (`platform_run_metrics`)	Email on a missed freshness target or a failed run
Status for each pipeline (last good run)	The platform status page (`status.json`)	Shown on the page; email on first failure
Stream lag (median and 95th percentile) and messages per second, per live session	Project metrics, one row per session	Reviewed at the end of each session
Model calls: model, prompt version, tokens, latency, cost	The call log in S3 (platform decision)	Daily cap and monthly limit (section 4)
Agent runs	AWS X-Ray traces	Reviewed when the agent's proposals change
Review queue size and approval rate	Project metrics (ADR 0018, proposed)	Reviewed weekly once the queue exists

Each real incident gets one record: what broke, the cause and the fix (platform step 0.5). Incident records go in the platform incident table.

4. Cost limits

Item	Limit	Source
Spark jobs	Maximum rows read and written, and maximum runtime. Over either, stop before publishing and open an incident.	Numbers open until measured (OQ-1, OQ-23)
Session box	4-hour session limit, idle shutdown, 2-hour nightly maximum	Platform
Snowflake	3 credits a month; time limit on every query	Platform
Model API	$15 a month, a daily cap, and a token limit per answer	Platform (model API decision)
Global Fishing Watch API	50,000 requests a day and 1,500,000 a month per user; up to five tokens per user. A request over the limit returns HTTP 429 and blocks the user for 24 hours (daily) or 30 days (monthly).	Provider's license page, checked Oct 9, 2026
S3	Clean-up rules from the platform (dev 14 days, temporary 7 days, failed uploads 7 days, pipeline logs 90 days, security logs 1 year, old versions 30 days) and a size alarm	Platform
DynamoDB (review queue)	Provisioned within the free tier; point-in-time recovery on	Platform
Phase C cluster	"A few dollars" for one month on 3 to 4 nodes. The dollar limit is open (OQ-23).	Build plan wording only

The nightly run must stay under the GFW daily limit. The request count per run is set when the first real report is measured (OQ-6).

5. Runbooks

Runbooks are written for real failures only, one per failure, after it happens (platform rule 12). Project 3 has none yet. Until a real incident occurs, these are the failures to expect and where the first checks go:

Failure	First check	Platform runbook
Session box does not start at 10:00 UTC	Schedule and spot capacity	PRB-1
Box stops at the 2-hour nightly maximum	Run metrics: which task was still running	PRB-2
Spot server reclaimed mid-run	Captures and staging left in S3; the next run replays them	PRB-3
Key exposed (pasted in a commit or posted)	Rotate the key in the provider, then in Parameter Store	PRB-5
Budget action fires	Which service spent the money, and the cost guard that should have stopped it	PRB-7

Project-specific runbooks to write once a real failure happens: an AISStream connection that refuses a third connection; a Global Fishing Watch 429; a buoy file whose stations change; a classifier export that does not match the copy in the repo.

6. Backups and recovery

- Published Parquet is replaced only after the publish gate passes. A failed run leaves the old data in place.
- DynamoDB point-in-time recovery is on for the review queue, so a bad write or deletion can be undone.
- Raw captures and source files are immutable (platform rule 1). Public archives are kept as a manifest, not a copy, where the provider never changes them.
- Airflow's database is saved to S3 after each run (platform decision).

7. Routine checks

When	Check
Each morning after the nightly run	Status page: each pipeline's last good run; any late notice
Each week	Weekly smoke test result; spend against the model and Snowflake limits; GFW request count against the limit
Each session	Stream lag and messages per second recorded; the session box shut down
Each month	Storage size against the alarm; the review queue size and approval rate

Part 5: Model card

`Ocean Map/model-card.md`

Project 3: Ocean map. Model card

Field	Value
Status	Draft for John's review (not yet approved). Nothing is built, so no measured results exist yet.
Owner	John
Last updated	Oct 9, 2026
Related	design-doc.md (sections 7.1, 8.5, 8.6, 13), requirements.md (sections 6 and 8), adrs/0012, adrs/0013, adrs/0014, adrs/0018, adrs/0020, platform ADR 0014

The platform decided the model provider, the cheapest-model rule, the spend limit and the call log. This card covers the parts of Project 3 that use models.

1. What uses a model

Part	Kind	Role	Model	Phase
Vessel matching, candidate pairs	Embeddings	Find candidate vessel pairs for comparison	Not chosen (OQ-16)	B
Vessel matching, ambiguous pairs	Language model, batch API	Decide ambiguous pairs, with a reason and a confidence	The cheapest model under the platform rule; the exact model is recorded when it is chosen (OQ-16)	B
Vessel matching, classifier	Small trained classifier	Score pairs without model calls, trained on the language model's decisions and John's labels	Not chosen (OQ-18)	B
Buoy fault flags	Rules plus scikit-learn	Flag stuck, spiking or silent buoys	Not chosen; thresholds open (OQ-20)	B
Data-quality agent	Language model with tools	Read failed checks and low-confidence matches and propose a fix	The cheapest model, through the platform's agent foundations	B
Clusters and location briefings	Language model	Phase C only: vessel activity clusters labelled as hypotheses; batch briefings	Not chosen	C (design written later, ADR 0019)

2. Intended use

- Propose vessel matches for John to decide. A match is not applied until John approves it.
- Propose data-quality fixes (merge two vessels, mark a buoy faulty, open an issue) as items in the review queue. Nothing changes until John approves.
- Compare three matching versions (rules only; rules plus language model; classifier) and report which one is best on the same hand-labelled pairs.
- Explain a match with the reason and confidence the model gave, and show the evidence next to it.

3. Not intended for

- Writing text that the public reads. There is no generated text on any page. Buoy flags show the rule that fired and the readings (QR-25; design decision "Cut: LLM-written buoy notes", ADR 0014).
- Changing data directly. The agent's only write is `create_proposal` (ADR 0013). It cannot write to the lake, the contract tables or the review decisions.
- Making safety or navigation decisions. The GEBCO grid itself is not for navigation (see security-and-privacy.md), and nothing here adds to that use.

- Naming people. Vessel names and MMSI are internal. The model sees ship data, not personal data, and its output is not shown publicly (ADR 0008).
- Answering questions that a query did not return. Location briefings in Phase C must reject any number a query did not return (build plan, Phase C).

4. Evaluation

Check	What it measures	Data	Status
Matching comparison (step 3B.4)	Precision, recall, cost per 1,000 pairs and review-queue volume for all three versions	200 hand-labelled pairs, held out from training and prompts (ADR 0020)	Designed; not run
Duplicate rate (step 3B.3)	Duplicate rate before and after identity resolution	The same 200 pairs	Designed; definition open (OQ-17)
Evals in CI (step 3B.10, proposed)	Agent answers graded on tool choice, numbers and refusal; red-team prompts, including instructions hidden in source text	Question set with known answers, and a red-team set	Proposed (ADR 0018); sizes and thresholds open
Fault rules (step 3B.6)	Rule flags against known good and known faulty buoy histories	Buoy history	Designed; thresholds open (OQ-20)
Proposal approval rate (step 3B.5)	Share of agent proposals John approves	Logged decisions	Designed

The comparison includes a rules-only baseline, so any gain from the language model is measured against it. The pairs used for evaluation are never used to train the classifier or as examples in prompts (ADR 0020, default; John's approval requested under OQ-30).

5. Known limits and failure cases

No results exist yet, so no failure rate can be stated. These are the failure cases the design expects and will test:

- A language model that proposes a match for two ships that never met. The impossible-speed check (step 3B.3) is a rule, and it runs before the model's proposal is shown.
- Instructions inside a vessel name or an AIS message that try to change the agent's behaviour. Source text is treated as data (platform rule 13), and the red-team set tests it.
- A classifier that agrees with the language model's mistakes, because it was trained on those decisions. The held-out pairs are the check.
- Buoy faults that the rules miss, or that the rules flag wrongly (for example a station that is silent for a legitimate reason).
- Batch results that arrive late. The nightly run does not wait indefinitely for a batch; a time budget is set when measured (ADR 0012).

6. Safety and privacy

- No personal data is sent to a model on purpose. Vessel data is ship data; prompts do not include the owner's identity.
- Keys for the model API are in Parameter Store, read by role, and never in code or in the browser (see security-and-privacy.md).
- Every model call is logged to the call log with the model, prompt version, tokens, latency and cost (platform decision). Traces for agent runs go to AWS X-Ray.
- A daily cap and a monthly limit of $15 apply to the model API, set in the provider account before the first call (platform decision).
- The agent runs with least privilege and through allowlisted tools (platform rule 13).

7. Who checks the model

John approves each proposal the agent makes and each classifier version that replaces the current one. A new model version or a new prompt version is recorded in the changelog and re-evaluated on the held-out pairs before it is used.

Part 6: Security and privacy

`Ocean Map/security-and-privacy.md`

Project 3: Ocean map. Security and privacy

Field	Value
Status	Draft for John's review (not yet approved)
Owner	John
Last updated	Oct 9, 2026
Related	design-doc.md (sections 12.1, 12.2, 12.6), adrs/0008, adrs/0013, adrs/0015, adrs/0022, adrs/0023, platform security-and-privacy.md

The platform's security rules (keys in Parameter Store, environment separation, Terraform state, least-privilege roles) apply here. This document covers what is specific to Project 3.

1. What the project holds

The project collects no visitor data. It holds:

Data	Where	Personal?	Shown publicly?
Buoy readings and station metadata	S3 and Iceberg tables	No	Latest reading and time, with credit
GEBCO grid cut, and mapped and unmapped flags per cell	S3	No	Unmapped share per cell
Global Fishing Watch per-cell counts	S3	No	Counts with the "possible dark vessels" label
AIS messages (MMSI, ship name, position)	Kafka, S3 and Iceberg	Ship identifiers; a ship's name or MMSI can identify its owner	No. Maps and counts only (ADR 0008).
Vessel golden record and matching pairs	Iceberg (SCD Type 2)	Same as above	No
Review queue and John's decisions	DynamoDB	No (John's decisions only)	Owner only, behind one API key (ADR 0015)
Request logs for the lookup and review routes	CloudFront and API Gateway logs	IP addresses may appear in them	No. Addresses are removed (ADR 0022).
Model call log	S3 (platform)	No; prompts contain vessel data, not people	No

Deletion periods. Raw AIS captures and temporary files follow the platform clean-up rules (temporary 7 days; dev 14 days; pipeline logs 90 days; security logs 1 year; old versions 30 days). Review-queue rows are kept with point-in-time recovery on; they are not deleted on a schedule. The request logs keep no addresses beyond the short window in ADR 0022.

2. Keys and secrets

Key	Used by	Stored in	Browser?
Global Fishing Watch API key	Nightly run and tests	Parameter Store (SecureString), read by role	Never
AISStream API key	Capture task and live session producer, server side only	Parameter Store, read by role	Never (ADR 0008)
Model API key	Batch matching, data-quality agent	Parameter Store, read by role	Never

Key	Used by	Stored in	Browser?
Owner API key for the review routes	Review page	Parameter Store. How it reaches the page is open (OQ-31).	Not decided. Must not be stored in code the browser downloads.
Databricks and Snowflake credentials (Phase B)	Export and marts	Parameter Store	Never

Keys never appear in code, in GitHub, or in the Terraform state. gitleaks and Trivy run in CI (platform step 0.1).

Key exposure notice. A third-party web page was found that publishes a key that looks like an AISStream key, in plain text. The key is not reproduced here. John should check whether it is his key. If it is, he should rotate it in AISStream and update Parameter Store. This is an action for John, not a document change.

3. Roles

- One least-privilege role per pipeline (platform rule 13).
- The capture task role may read its key and write only to the capture prefix.
- The data-quality agent's role may read through allowlisted tools and call `create_proposal` only. It cannot write to the lake, the contract tables or the review decisions (ADR 0013).
- The owner-only review routes accept only the owner's key. The review page is not linked from the public site (ADR 0015).
- Dev roles cannot write live data (platform environment rule).

4. Threats

Threat	Impact	Control	Status
A key is pasted into code, a commit or a web page	Key used by others; cost and abuse	Parameter Store only; gitleaks and Trivy in CI; rotation runbook (PRB-5)	Designed
Vessel names or MMSI shown publicly	Privacy of ship owners; a legal risk while the AISStream terms are unknown	Maps and counts only (ADR 0008)	Designed; terms open (OQ-3)
Source text (vessel names, AIS messages) carries instructions to the agent	The agent acts on text it was given	Source text is data, never instructions (platform rule 13); red-team test set (ADR 0018, proposed)	Designed; tests not written
The review page is found and used by someone other than the owner	Unauthorised decisions	Keyed REST routes; page not linked; the key delivery method is open (OQ-31)	Partly designed
Lookup abused for cost (many requests)	Cost and availability	Throttling and reserved concurrency; limits open (OQ-10)	Open
IP addresses in logs	Personal data kept beyond need	Addresses removed before storage beyond the short window (ADR 0022)	Designed
Global Fishing Watch rate limit exceeded	429 block for 24 hours or 30 days	Request count per run kept under the daily limit (operations.md section 4)	Designed; count open (OQ-6)
Buoy, GFW or NOAA data used in a way the licence does not allow	Breach of terms	Licence register (section 5); ADR 0023	Partly verified

5. Licence and credit register

Each entry is checked against the provider's own page, as ADR 0023 requires. Status is one of:

- **Verified**: the terms were read on the provider's page.
- **Secondary only**: the statement comes from a third-party page, not the provider's.
- **Unverified**: the provider's page could not be read, or no statement was found on it.

Source	Terms as found	Credit required	Status	Action
Global Fishing Watch APIs (4Wings, Phase A)	Licensed CC BY-NC 4.0, "available for noncommercial use only". Use is allowed only when not intended "primarily or materially" for commercial advantage or monetary compensation. GFW may revoke use that breaks these terms. Source: GFW license and rate limits page (fetched Oct 9, 2026).	Yes. Option: "Powered by Global Fishing Watch.", linked to globalfishingwatch.org. Longer citation options are on the same page.	**Verified** (primary page)	Keep the page non-commercial. Use the "Powered by" text on the page.
GEBCO Grid (Phase A)	"The GEBCO Grid is placed in the public domain and may be used free of charge." Users must acknowledge the source, must not suggest official status or endorsement, and must not misrepresent the grid. Not for navigation or safety at sea. Accuracy is not guaranteed. Source: GEBCO terms of use page (fetched Oct 9, 2026).	Yes. Suitable wording is "given in the documentation that accompanies The GEBCO Grid", which was not read.	**Verified** for the terms. **Unverified** for the credit wording.	Read the documentation that comes with the grid download and copy its credit line. Check the DOI for the version used. An older BODC-hosted copy mentions copyright and database rights; check which text applies to the download.
NOAA NDBC buoy data (Phase A)	No data policy or credit statement found on the NDBC home page. Secondary sources say the data is public domain (one API vendor) and CC0 (a 2015 open-data catalogue record). Not confirmed on NDBC's own page.	Unknown	**Unverified** (NDBC page); secondary only	Find NDBC's own data policy or "about" page, or ask NDBC. Until then the page's credit line is generic and John decides whether Phase A may publish (see section 6).
MarineCadastre AIS (Phase C only)	Stated as CC0 in a NOAA OCM GitHub repository (secondary). The MarineCadastre hub page (fetched Oct 9, 2026) gave no licence text.	Unknown	**Unverified** (primary); secondary only	Check the licence on the specific product before Phase C uses it.
AISStream (Phase B)	No terms page found in search. The provider's own site returned HTTP 403 when fetched.	Unknown	**Unverified**	John will ask AISStream for its terms (OQ-3). Until they answer, only maps and counts with credit are shown (ADR 0008).
Model API provider	Terms govern the model calls; the platform sets the spend limit and the cheapest model (platform model API decision).	Not applicable to the page	Not checked in this revision	Check the provider's terms when the first call is made.
MMSI country prefix table	No source chosen	Not known	Open (OQ-15, ADR 0017)	None until a source is chosen
OpenCellID and others from Project 1	Not used in Project 3	Not applicable	Not applicable	None

6. Decisions for John

- Whether Phase A may publish derived NDBC data before its licence is verified. Recommendation: yes, with the generic credit line, once NDBC's own policy page is read. Nothing is published from NDBC until that page is read.
- Whether to ask GFW for a custom licence if the project ever earns money. The current terms allow only non-commercial use.

7. Platform level

The platform's security and privacy rules are in Platform/security-and-privacy.md. Project 3 relies on the platform's keys, roles, environment separation, Terraform state and clean-up rules without repeating them.

Part 7: Architecture decision records

Ocean Map/adrs/0001-phase-a-region-scope.md

0001. Phase A covers one region

- Status: Proposed (the region itself is not chosen; see open question OQ-2 in the design doc)
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

The build plan sets Phase A scope as "one region (for example the Gulf of Mexico)". The general rule for Phase A is the smallest version that works end to end, on a small area or sample, using the same stack as the full version. Global GEBCO coverage is deliberately left to Phase C. The AIS message rate for a region is not yet measured (OQ-1), and Global Fishing Watch reports are limited to one at a time per user.

Decision

Phase A and Phase B cover a single region. The Gulf of Mexico is the plan's example and is the working assumption in diagrams and text only as an example. The final region and its bounding box are chosen by John and recorded here, replacing this proposal, before step 3A.2 starts.

Alternatives considered

- Global coverage in Phase A: rejected by the plan; global GEBCO coverage is Phase C.
- Several regions in Phase A: not proposed; the plan names one region.
- A different single region: possible; this is the open choice.

Consequences

- Row counts, report sizes and cost limits are set for one region.
- The region's bounding box is an input to the GEBCO cut, the Global Fishing Watch requests, the NDBC station list and the AISStream subscription.
- Moving to another region or to the globe changes data volumes and must update the Spark job limits.

Ocean Map/adrs/0003-mapped-seafloor-rule-gebco-source-type-codes.md

0003. Mapped seafloor rule from GEBCO source-type codes

- Status: Accepted for the code meanings (recorded in the build plan risks table and the Documentation Plan). The step that combines pixel codes into a cell flag is open (OQ-4).
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

Step 3A.2 uses GEBCO's source-type grid to flag cells as mapped or unmapped and says to "write the rule down". GEBCO type-identifier codes: 10 to 17 are direct measurements (echo sounders, seismic, lidar, charts, optical); 40 to 46 are interpolated or predicted; 70 to 72 are unknown; 0 is land. The first-run check confirms that the codes are read correctly for the region.

Decision

- Codes 10 to 17 count as mapped.

- Codes 40 to 46 (interpolated or predicted) and 70 to 72 (unknown) do not count as mapped, so they count as unmapped. The plan states that 10 to 17 "count as mapped" and lists the others as not measured; treating all non-measured ocean codes as unmapped is the reading used here.
- Code 0 is land. Land cells are not ocean cells and receive no flag.
- The page states which codes are counted as mapped, so a visitor knows "unmapped" means "not directly measured".

Alternatives considered

- Count interpolated or predicted codes (40 to 46) as mapped: rejected because they are not direct measurements and would understate the unmapped share.
- Treat unknown codes (70 to 72) as a separate third category on the page: not in the plan; could be added if John wants it.

Consequences

- The unmapped share is a statement about direct measurement, not about whether any depth value exists.
- A unit test with a sample tile of known codes protects the rule (QR-12).
- OQ-4 must be answered before 3A.2 is built: the rule for combining pixel codes into one H3 resolution 8 cell.

`Ocean Map/adrs/0006-live-ais-stream-sessions-and-nightly-replay.md`

0006. Live AIS through Kafka during sessions, daily Fargate capture, nightly replay

- Status: Accepted (recorded in step 3B.1 and the decision "Live stream"; "Ship captures"; "Always-on parts")
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

Kafka runs only while the session box is on. Lambda and Fargate write only to always-on storage (S3, DynamoDB, CloudWatch, X-Ray) so that everything stays on AWS at about $0 and a small always-on server is avoided, because it would break the budget (decision "Always-on parts"). Captures are written straight to S3 so that replay works after the server is gone (decision "Ship captures"). The nightly run uses spot only and can be skipped; the next run catches up safely.

Decision

- During sessions, an AISStream websocket client on the session box produces messages to Kafka, and a Structured Streaming job consumes them and writes Iceberg.
- Outside sessions, a daily one-hour Fargate capture writes raw messages to S3 only.
- The nightly run replays the day's captures through Kafka with the same streaming job, using Spark's availableNow trigger, so the job processes everything new and stops.
- The same job code serves live and replay modes; only the trigger and source differ.
- The job parses the semi-structured messages, handles late and duplicate events and writes to Iceberg.

Alternatives considered

- A small always-on server for ingest: rejected; breaks the budget.
- Fargate producing to Kafka: rejected; Kafka is off outside sessions.
- Capturing only during sessions: rejected; leaves the data to depend on John's work hours.
- Separate code for live and replay: rejected; two code paths would drift.

Consequences

- Captures are the only record of hours when no session ran; a missed capture is a permanent gap (AIS cannot be fetched after the fact).
- Replay must be idempotent (ADR 0007) because a live session and the capture can observe the same hour, and a skipped night leads to a double replay later.
- Session lag and messages per second are measured in live mode only.
- Message rate (OQ-1) and capture file format (OQ-12) decide capture storage and job limits.

`Ocean Map/adrs/0007-event-identity-and-deduplication.md`

0007. Event identity and deduplication for the AIS stream

- Status: Proposed (the approach follows platform standards and Project 1's streaming pattern; the key and watermark values are open: OQ-14)
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

Engineering standards 2 and 3 require event time and a source key on every record, and idempotent, atomic tasks. The 3B.1 Done when requires that replaying a capture twice writes each event once. Step 3B.1 requires the job to handle late and duplicate events. Project 1's step 1B.2 uses an explicit schema, checkpoints on S3, a watermark and dropDuplicatesWithinWatermark with the availableNow trigger.

Decision

- Every AIS record carries event time (UTC) and a source key.
- The stream job deduplicates by source key within a watermark using dropDuplicatesWithinWatermark, with an explicit schema and checkpoints on S3, as in Project 1's step 1B.2.
- Writes to Iceberg are idempotent so a replayed capture adds no rows already present.
- What the source key consists of, and how long the watermark is, are not decided (OQ-14).

Alternatives considered

- Deduplicate in dbt after landing: leaves duplicates in the lake table and breaks the "each event once" check at the table level.
- Append without keys: violates standard 3.
- A random id per message: makes duplicates indistinguishable.

Consequences

- The key must be chosen so that two genuine messages are never merged and one message captured twice always is; this needs a look at real messages (OQ-13, OQ-14).
- Messages that arrive later than the watermark may be dropped or duplicated; the rate is recorded once measured.
- The CI test replays a sample capture twice and compares rows.

`Ocean Map/adrs/0008-aisstream-usage-and-display-limits.md`

0008. AISStream usage and display limits

- Status: Accepted. Connection rules recorded in the Phase B first-run check and the risks table "Live AIS terms". Display scope decided by John, Oct 9, 2026: maps and counts only. John will ask AISStream about their terms (OQ-3, OQ-24).
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

AISStream.io is free, allows three connections and has no terms page. The Phase B first-run check says it must not be opened from a browser. The risks table says: show only derived maps and counts, with credit. Step 3B.1 also says to confirm its terms. The project summary says the lookup shows "which vessels are active".

Decision

- The websocket is opened only from server-side code: the Fargate capture task and the producer on the session box. Never from a browser.
- At most three connections are used at once. The design uses two at most (one live producer in a session, one daily capture); any other use, for example the weekly smoke test (which pulls one small sample from every source), is brief and counted.
- The AISStream key is stored in Parameter Store and read by role.
- Public output is limited to maps and counts with credit to AISStream. Vessel names and MMSI are internal and never shown publicly (decided by John, Oct 9, 2026).
- John will ask AISStream about their terms. The display rule is loosened only after their answer.
- Terms are confirmed before publishing anything derived from AISStream (3B.1: "confirm its terms").

Alternatives considered

- Browser connection: rejected; exposes the key and violates the first-run check.
- Showing vessel names and tracks publicly: not allowed under the current decision; needs terms confirmation.
- Another AIS source: not in the plan for Phase B; MarineCadastre is used in Phase C.

Consequences

- The "which vessels are active" part of the project summary is delivered as counts and maps for now.
- The vessel identity table stays internal; the review queue page shows vessel names to John only (owner only behind one API key, ADR 0015).
- Three connections must be accounted for when the smoke test, a live session and the capture overlap.

`Ocean Map/adrs/0009-gfw-non-commercial-use-and-request-handling.md`

0009. Global Fishing Watch: non-commercial use and one-request-at-a-time handling

- Status: Accepted (facts from step 3A.3, plan assumptions and the Documentation Plan "Global Fishing Watch fields")
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

Global Fishing Watch requires an API key and credit, and its use is non-commercial. The 4Wings report endpoint returns CSV or JSON with date, lat, lon and hours or detections. SAR can be filtered to matched or unmatched. One report can run at a time per user, and a report over 100 seconds is recovered through last-report. The plan states licensing must be checked before charging for any project.

Decision

- The Global Fishing Watch credit appears on the page and in the README. The project is non-commercial.
- The ingest task requests one 4Wings report at a time; the Airflow task runs with a concurrency of one for this source (a pool of one slot).
- A report that exceeds 100 seconds is recovered through last-report rather than requested again.

- The data is gridded; per-cell counts are stored, not per-vessel rows.
- The API key lives in Parameter Store.

Alternatives considered

- Parallel report requests: not allowed by the service (one at a time per user).
- Re-requesting a slow report: wastes credit and conflicts with the one-at-a-time rule.
- Commercial use: not available; would need a license check first.

Consequences

- Report time bounds the length of the ingest task; the number of reports per run follows from OQ-6.
- The project cannot be monetized without a licensing review of Global Fishing Watch (plan assumptions list it as non-commercial).
- Exact credit wording is open (OQ-24).

`Ocean Map/adrs/0011-vessel-identity-golden-record-and-review-queue.md`

0011. Vessel identity: golden record, human review queue, labels as SCD Type 2

- Status: Accepted (step 3B.3; access decided by John, Oct 9, 2026; decision "Visitor data"; standard 8). Parameters are open (OQ-15, OQ-16, OQ-17).
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

AIS messages carry an MMSI and ship name (MetaData) and, in typed messages, further fields. One MMSI can be used by two ships at once, names vary and IMO numbers have check digits. Step 3B.3 asks for a golden vessel record: validate MMSI country prefixes and IMO check digits, catch one MMSI used by two ships at once by impossible speed between positions, match name variants. Low-confidence matches go to a review queue in DynamoDB and a React page; John's decisions are stored as labels and applied as SCD Type 2 rows. DynamoDB is the always-on store for writable, queue-like data while the box is off.

Decision

- Build a golden vessel table from AIS observations with rule-based validation and matching.
- Route pairs by confidence: confident pairs are accepted, low-confidence pairs go to the review queue in DynamoDB.
- Review decisions are stored as labels; the nightly run copies the queue to the lake and applies labels as SCD Type 2 rows.
- Report the duplicate rate before and after and precision and recall on 200 hand-labeled pairs.
- The review page is a React page backed by Lambda. Decided by John, Oct 9, 2026: owner only, behind one API key in Parameter Store, not linked from the public site (ADR 0015).

Alternatives considered

- Fully automatic matching without review: rejected by the plan; low-confidence matches need a human.
- Review in a spreadsheet or GitHub issues: not in the plan; the React page and DynamoDB queue are the recorded design.
- Overwrite the golden record in place: rejected by standard 8; history stays in rows.

Consequences

- The queue is a write path; it is owner only behind one API key (ADR 0015), which must be in place before the page is built.
- An MMSI prefix reference table is required but is not in the plan's source list (OQ-15).
- IMO numbers and other static fields must exist in the AISStream messages used (OQ-13).
- The 200 evaluation pairs must be kept separate from training data (OQ-17).
- Labels become training data for ADR 0012 version 3.

`Ocean Map/adrs/0012-ai-assisted-matching-three-version-comparison.md`

0012. AI-assisted vessel matching: compare three versions, then choose

- Status: Proposed, pending measurement. The comparison design is recorded (step 3B.4); the chosen version is not known until measured. This record is replaced (or superseded) by the decision record that 3B.4 requires.
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

Step 3B.4 compares three versions: (1) rules only; (2) rules plus AI, where embeddings find candidate pairs and an LLM decides ambiguous ones through the batch API with a reason and confidence; (3) a small classifier trained on the LLM's decisions and John's labels, so the pipeline runs without model calls. AI placement decision: AI ship matching with a cheaper trained copy. Only the cheapest model is used for now; the monthly limit is $15; generated content is made in batch and cached. LanceDB files in S3 hold embeddings. Metrics reported: precision, recall, cost per 1,000 pairs and review-queue volume.

Decision

Implement all three versions behind one interface, run them on the same hand-labeled pairs, report precision, recall, cost per 1,000 pairs and review-queue volume for each, and choose the version from those numbers. The choice, with its reasons, replaces this record.

Alternatives considered

- Choose version 2 up front: rejected; the plan requires the comparison.
- Call a model for every pair: rejected; cost and the $15 limit.
- Fine-tune a large model: not in scope; the plan says a small classifier.
- Use a stronger model: not allowed for now; only the cheapest model is used.

Consequences

- The evaluation pairs and training data must be disjoint (OQ-17).
- The classifier is trained on Databricks Free Edition with manual export (download results and model, commit to GitHub), as Projects 1 and 5 (decided by John, Oct 9, 2026). The pipeline scores with the copy in the repo, so it never depends on Databricks being up. Whether the export can be automated stays an open test (OQ-18).
- Whichever version wins, the others stay as evaluation baselines in CI.
- The batch API delays results; the nightly run must not wait indefinitely on a batch (a time budget is set when measured).

Ocean Map/adrs/0013-data-quality-agent-proposal-only.md

0013. Data-quality agent can only create proposals

- Status: Accepted (step 3B.5; engineering standard 13 (least privilege for agents); AI placement decision)
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

Agents that touch real systems are the main safety risk in the AI layer. Standard 13: least privilege, read-only roles and allowlists for agents, scoped MCP tools, write actions only through human approval, every key in Parameter Store, and source text treated as data, never as instructions. Step 3B.5: the agent reads failed checks and low-confidence matches and proposes fixes (merge two vessels, mark a buoy faulty, open an issue). Its only write tool creates a proposal in the review queue; nothing changes until John approves. Agent queries use defined metrics first, with raw SQL as a logged fallback. Tracing goes to X-Ray.

Decision

- The agent's tools are read-only except `create_proposal`, which writes one item to the review queue.
- The agent's IAM role has no write access to lake tables, contract tables, GitHub or any other store.
- Proposals, John's decisions and the approval rate are logged.
- Text found in source data (vessel names, check messages) is data, never instructions.
- Only the cheapest model is used, with structured output.

Alternatives considered

- Direct writes with an undo: rejected; the plan requires no direct write access.
- Opening GitHub issues directly: rejected; "open an issue" is itself a proposal that John approves first.
- No agent: rejected by the plan; the technology table lists MCP servers and agents for Project 3 phase B.

Consequences

- Where the agent runs and how approved proposals are applied are open (OQ-19).
- A red-team set (instructions hidden in source text, write requests) tests the permission boundary.
- Approval rate measures usefulness; low approval would argue for removing or changing the agent.

Ocean Map/adrs/0014-buoy-fault-detection-rules-and-scikit-learn.md

0014. Buoy fault detection with rules and scikit-learn, no generated notes

- Status: Accepted for the approach (step 3B.6; decision "Cut: LLM-written buoy notes"). Thresholds and model are open (OQ-20).
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

Step 3B.6: rules plus scikit-learn flag stuck, spiking or silent buoys. The page shows the flag, the rule that fired and the readings, with no generated note. LLM-written buoy notes were cut because they were AI for show; multi-agent work moved to the on-call agent. Buoy history comes from the 3B.2 backfill.

Decision

- Fault flags come from deterministic rules for stuck, spiking and silent readings, with a scikit-learn model as a second signal.
- Every flag records the rule that fired, and the page shows the flag, the rule and the readings.
- No model call generates text for buoy flags.

Alternatives considered

- LLM-written notes on each flag: cut by decision.
- A model without rules: rejected; the plan requires showing the rule that fired.
- Rules only: rejected; the plan lists scikit-learn for this step.

Consequences

- Flags are explainable and cost nothing in model calls.
- Rule thresholds and the model's role relative to rules need experiments on history (OQ-20).
- The data-quality agent can propose marking a buoy faulty from a flag (ADR 0013), but a flag itself is a pipeline output, not an agent action.

`Ocean Map/adrs/0015-review-queue-owner-only-api-key.md`

0015. Review queue writes: owner only, one API key, not linked from the public site

- Status: Accepted. Decided by John, Oct 9, 2026. The key delivery mechanism is open (OQ-31).
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

The review queue (ADR 0011) holds low-confidence vessel matches and data-quality agent proposals, and John's decisions on them become labels and SCD Type 2 rows. A write path on a public site needs access control, and the build plan described a React page without saying how access is limited. The REST type of API Gateway is the type that supports API keys. Every key is stored in Parameter Store (decision "Key storage"). Vessel names and MMSI are internal (ADR 0008).

Decision

- Only the owner can write review-queue decisions.
- The review routes use the keyed REST type of API Gateway, behind one API key held in Parameter Store.
- The review page is not linked from the public site.
- The data-quality agent does not use these routes. Its only write is `create_proposal` into the queue, with a role limited to that action (ADR 0013).

Alternatives considered

- Open routes with the page linked from the site: rejected; a write path would be public.
- A separate login service: not chosen; one owner and one key are enough.
- HTTP type for these routes: rejected; the build plan says only the REST type supports API keys.

Consequences

- The key must be in place before the page is built (requirement FR-58).
- Key rotation is not decided and is settled at build.

- How the key reaches the page is not decided (OQ-31). A key placed in code the browser downloads is visible to anyone who opens the page, so the mechanism must be decided before the review page is built. One option is that the owner types the key into the page at the start of each session and the page never stores it in code.
- Public pages never call these routes.

`Ocean Map/adrs/0018-metrics-evals-tracing-and-mcp-step-list.md`

0018. Metrics, evals, tracing and MCP: proposed step list for approval

- Status: Proposed, awaiting John's approval. Nothing in the list is approved or built. The metric set, eval sizes and thresholds are open (OQ-21, OQ-30).
- Date: Oct 9, 2026
- Project: 3 Ocean map

Context

The technology table lists MetricFlow, MCP servers and agents, and LLM evals, red-team tests and tracing for Project 3 Phase B. No build step defines them, so the design had no tasks or done-when checks for them. On Oct 9, 2026 John decided that a short proposed step list is written for his approval.

Decision

The four proposed steps in requirements.md section 8 are the proposal:

- 3B.7 Metrics: a small MetricFlow metric set on the Snowflake marts; one metric must give the same value from Snowflake and DuckDB.
- 3B.8 Call log and tracing: the platform call log for the batch matching calls and the data-quality agent; agent runs traced to AWS X-Ray.
- 3B.9 MCP server: metrics exposed as read-only tools; raw SQL only as a read-only, allowlisted fallback; the data-quality agent's only write stays `create_proposal` (ADR 0013).
- 3B.10 Evals in CI: a question set with known answers and a red-team set; CI fails below the thresholds.

Sizes marked "mirrors Project 1" copy Project 1 steps 1B.5 and 1B.7 and need approval. The step list does not change the plan until John approves it.

Alternatives considered

- Leave the steps out: rejected; John asked for the list.
- Write full task plans and numbers now: rejected. The numbers depend on measurement and are not approved.
- Copy Project 1's steps unchanged: rejected as a default. The sizes need John's approval first.

Consequences

- Phase B cannot be called complete until the approved steps have their done-when checks.
- The metric set is John's choice; the candidates in the list are only candidates.
- The thresholds for evals are set after approval, from measured results.

Part 8: Data contracts

Ocean Map/contracts/source-aisstream.yaml

```
# ODCS data contract: source. AISStream.io live AIS messages and raw capture files (step 3B.1)
apiVersion: v3.0.2
kind: DataContract
id: project3-source-aisstream
name: AISStream live AIS messages
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Live vessel positions and static data for the region, streamed through Kafka during sessions and captured
daily for nightly replay.
  limitations: >
    Free key; three connections; must not be opened from a browser; no terms page, so only maps and counts are shown,
with credit; vessel names and MMSI are internal. John will ask AISStream about their terms and the rule is loosened
only after their answer (ADR 0008, decided by John, Oct 9, 2026).
    Message rate for the region is unknown (design doc OQ-1). Which message types and fields are used, and which carry
IMO number
    and other static fields, are not confirmed (OQ-13). The raw capture file format is open (OQ-12).
  usage: Written to S3 by C9 (Fargate capture) and to Kafka by C10 (live producer); parsed by C11.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: aisstream-websocket
    type: custom
    description: AISStream websocket. Subscribe with an API key, bounding boxes and optional MMSI and message-type
filters. Server side only.
schema:
  - name: aisstream_message
    description: One websocket message. 25 message types exist.
    properties:
      - name: MessageType
        logicalType: string
        required: true
        description: One of the 25 AISStream message types.
      - name: MetaData
        logicalType: object
        required: true
        description: Contains the MMSI, the ship name and the position (exact key names confirmed on first connection,
OQ-13).
      - name: Message
        logicalType: object
        required: true
        description: The typed message body; its shape depends on MessageType.
  - name: aisstream_capture_part
    description: A raw capture file written by the Fargate task. Format and part size are open (OQ-12).
    properties:
      - name: part_uri
        logicalType: string
        required: true
      - name: sha256
        logicalType: string
        required: true
        description: Hash recorded in the capture manifest (hash algorithm fixed by the platform raw-file standard).
      - name: first_event_time
        logicalType: timestamp
        required: false
      - name: last_event_time
        logicalType: timestamp
        required: false
quality:
  - type: custom
    engine: pipeline
    description: Messages that do not match the explicit schema are routed aside, not dropped silently (handling to be
defined with OQ-13).
slaProperties:
  - property: frequency
```

```
    value: 1
    unit: d
    description: One hour of capture per day; live sessions on demand.
customProperties:
  - property: connectionsAllowed
    value: 3
  - property: browserAccess
    value: "forbidden"
  - property: keyLocation
    value: "Parameter Store"
```

Ocean Map/contracts/source-gebco-grid.yaml

```
# ODCS data contract: source. GEBCO 2026 grid and source-type identifier grid (step 3A.2)
apiVersion: v3.0.2
kind: DataContract
id: project3-source-gebco-grid
name: GEBCO 2026 grid and source-type identifier grid
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Seafloor elevation and the grid that says what kind of data supports each pixel; used to flag H3 resolution
8 cells mapped or unmapped.
  limitations: >
    A static public archive: the plan stores a manifest of URL, size and hash, not a full copy (platform rule 1). The
regional cut is a derived file.
    The file format, tile layout, download URL and license/credit wording are not recorded in the sources (design doc
OQ-24, OQ-25).
    Whether a 2026 source-type grid is published in the same form as earlier releases is confirmed by the first-run
check
    "GEBCO's source-type codes are read correctly for the region".
  usage: Read by C2 (cut_region with xarray or rasterio). Phase C extends to the global grid.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: gebco
    type: custom
    description: GEBCO downloads. URL recorded in the manifest at first fetch.
schema:
  - name: gebco_grid
    description: 15 arc-second raster (resolution from platform rule 9, H3 resolution 8).
    properties:
      - name: latitude
        logicalType: number
        required: true
      - name: longitude
        logicalType: number
        required: true
      - name: elevation
        logicalType: number
        required: true
        description: Elevation/depth value of the pixel. Units and sign convention confirmed at first read.
  - name: gebco_source_type_grid
    description: Source-type identifier code per pixel.
    properties:
      - name: latitude
        logicalType: number
        required: true
      - name: longitude
        logicalType: number
        required: true
      - name: source_type_code
        logicalType: integer
        required: true
        description: >
          Codes 10 to 17 are direct measurements (echo sounders, seismic, lidar, charts, optical) and count as mapped.
          Codes 40 to 46 are interpolated or predicted. Codes 70 to 72 are unknown. Code 0 is land.
        quality:
          - type: custom
            engine: pipeline
```

```
            description: Values outside 0, 10 to 17, 40 to 46, 70 to 72 are reported (design doc OQ-4 for handling).
slaProperties:
  - property: frequency
    value: "on release"
    description: Re-read when a new GEBCO release or a larger region is chosen, not nightly.
customProperties:
  - property: rawRetention
    value: "Manifest of URL, size and hash (public archive that never changes)"
  - property: schemaDiffBeforeLoad
    value: true
```

Ocean Map/contracts/source-gfw-4wings.yaml

```
# ODCS data contract: source. Global Fishing Watch 4Wings report API (step 3A.3)
apiVersion: v3.0.2
kind: DataContract
id: project3-source-gfw-4wings
name: Global Fishing Watch 4Wings report
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Gridded AIS fishing presence and SAR detections without an AIS match ("possible dark vessels") for the
region.
  limitations: >
    Non-commercial; credit required (ADR 0009). The report returns CSV or JSON with date, lat, lon and hours or
detections.
    Dataset names, temporal resolution, grid resolution and data delay are not confirmed (design doc OQ-6).
    One report at a time per user; a report over 100 seconds is recovered through last-report.
    SAR detections with no AIS match are possible dark vessels, not confirmed ones (ADR 0004).
  usage: Read by C3 (gfw_fishing_dag) with the API key from Parameter Store.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: gfw-4wings
    type: custom
    description: Global Fishing Watch API, 4Wings report endpoint. Authenticated with an API key.
schema:
  - name: gfw_4wings_report_row
    description: One grid cell for one date and one layer.
    properties:
      - name: date
        logicalType: date
        required: true
        description: Report date (period granularity is open, OQ-6).
      - name: lat
        logicalType: number
        required: true
        description: Grid cell latitude as returned.
      - name: lon
        logicalType: number
        required: true
        description: Grid cell longitude as returned.
      - name: hours
        logicalType: number
        required: false
        description: Present for the AIS fishing layer.
      - name: detections
        logicalType: number
        required: false
        description: Present for the SAR layer, filtered to matched or unmatched. This project keeps unmatched.
quality:
  - type: custom
    engine: great-expectations
    description: Non-empty result for the region and interval; hours and detections are non-negative.
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Requested by the nightly run. Data delay from the provider is unknown (OQ-6).
customProperties:
```

```
  - property: concurrency
    value: "one report at a time per user"
  - property: timeoutRecovery
    value: "reports over 100 seconds recovered through last-report"
  - property: rawRetention
    value: "Raw responses immutable with hash and load time"
  - property: schemaDiffBeforeLoad
    value: true
```

Ocean Map/contracts/source-mmsi-prefix-reference.yaml

```
# ODCS data contract: source. MMSI country prefix reference (step 3B.3)
# NOTE: this source is NOT in the build plan's source list. It is required by "validate MMSI country prefixes".
# Its origin and license are open (design doc OQ-15).
apiVersion: v3.0.2
kind: DataContract
id: project3-source-mmsi-prefix-reference
name: MMSI country prefix reference table
version: 0.1.0
status: proposed
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Maps the leading digits of an MMSI to a country so that MMSI values can be validated.
  limitations: Source, license, update frequency and format are undecided (OQ-15).
  usage: Read by check_mmsi (component C14).
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: to-be-decided
    type: custom
    description: Undecided (OQ-15).
schema:
  - name: mmsi_prefix
    properties:
      - name: prefix
        logicalType: string
        required: true
        primaryKey: true
        description: Leading digits that identify a country.
      - name: country
        logicalType: string
        required: true
slaProperties:
  - property: frequency
    value: "on change"
    description: Re-read manually when the reference changes; cadence undecided.
```

Ocean Map/contracts/source-ndbc-history.yaml

```
# ODCS data contract: source. NOAA NDBC historical buoy data (step 3B.2)
apiVersion: v3.0.2
kind: DataContract
id: project3-source-ndbc-history
name: NOAA NDBC historical buoy readings
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Buoy history by year, past NDBC's 45-day realtime window, for buoy fault detection and the Snowflake marts.
  limitations: >
    File layout, years available per station and missing-value markers are not yet confirmed (design doc OQ-7, OQ-25).
    Missing stations and years are expected and are logged, not treated as errors.
  usage: Read by ndbc_history_backfill_dag (component C12), run only in work sessions (never in the nightly run).
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
```

```
servers:
  - server: ndbc-history
    type: custom
    description: NOAA NDBC historical files over HTTPS. URL pattern to be recorded at first load (OQ-7).
schema:
  - name: ndbc_history_file
    physicalName: per-station per-year file (layout to be confirmed, OQ-7)
    description: One file per station and year.
    properties:
      - name: station_id
        logicalType: string
        required: true
      - name: year
        logicalType: integer
        required: true
        description: Partition key; a rerun overwrites only this station-year partition.
      - name: observation_time
        logicalType: timestamp
        required: true
        description: Stored as UTC.
      - name: wave_height
        logicalType: number
        required: false
      - name: water_temperature
        logicalType: number
        required: false
      - name: other_measurements
        logicalType: object
        required: false
quality:
  - type: custom
    engine: pipeline
    description: Every missing station or year is written to the gaps log.
slaProperties:
  - property: frequency
    value: "on demand"
    description: Backfill runs by John in work sessions; a later periodic refresh is not defined in the plan.
customProperties:
  - property: rawRetention
    value: "Raw files immutable with source URL, fetch time and hash"
  - property: schemaDiffBeforeLoad
    value: true
```

Ocean Map/contracts/source-ndbc-realtime.yaml

```
# ODCS data contract: source. NOAA NDBC realtime buoy data (step 3A.1)
# Version of the standard is a placeholder until the platform contract template exists (design doc OQ-27).
apiVersion: v3.0.2
kind: DataContract
id: project3-source-ndbc-realtime
name: NOAA NDBC realtime buoy readings
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Latest buoy conditions (waves, water temperature) for the stations in the Phase A region, shown on the
lookup page as the nearest buoy's latest conditions.
  limitations: >
    NDBC realtime data covers a 45-day window; older data comes from the history source (source-ndbc-history.yaml).
    Exact file layout, column names, units and missing-value markers are not yet confirmed (design doc OQ-7) and are
fixed
    by the schema-diff step at first load. Terms and credit wording are open (OQ-24).
  usage: Read by the ndbc_realtime_dag Airflow task (component C1). Raw files are kept immutable with source URL, fetch
time and hash.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: ndbc-realtime
    type: custom
    description: NOAA NDBC realtime files over HTTPS. Exact URL pattern to be recorded at first load (OQ-7).
schema:
```

```
  - name: ndbc_realtime_file
    physicalName: per-station realtime file (layout to be confirmed, OQ-7)
    description: One file per station; each row is one observation time.
    properties:
      - name: station_id
        logicalType: string
        required: true
        description: NDBC station identifier. Part of the source key.
      - name: observation_time
        logicalType: timestamp
        required: true
        description: Time of observation. Source time zone is to be confirmed; stored as UTC (platform rule 15).
      - name: wave_height
        logicalType: number
        required: false
        description: Wave height. Unit and source column name confirmed at first load (OQ-7). Range check applies
(bounds open, OQ-7).
      - name: water_temperature
        logicalType: number
        required: false
        description: Water temperature. Unit and source column name confirmed at first load (OQ-7). Range check applies
(bounds open, OQ-7).
      - name: other_measurements
        logicalType: object
        required: false
        description: Any further measurements in the file. Which are shown on the page is open (OQ-7).
quality:
  - type: library
    metric: duplicateValues
    arguments:
      properties: [station_id, observation_time]
    mustBe: 0
    description: Source key (station and observation time) is unique after load.
  - type: custom
    engine: great-expectations
    description: Range checks on wave height and water temperature at the publish gate. Bounds are open (OQ-7).
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Pulled by the nightly run (10:00 UTC). The page shows an "as of" time and a late notice when the run
is late (ADR 0016).
  - property: freshness
    value: "open"
    description: Numeric freshness target is open (OQ-8).
customProperties:
  - property: rawRetention
    value: "Raw files immutable with source URL, fetch time and hash; live raw files Object Locked 30 days"
  - property: schemaDiffBeforeLoad
    value: true
  - property: weeklySmokeTest
    value: true
```

Ocean Map/contracts/source-ndbc-stations.yaml

```
# ODCS data contract: source. NOAA NDBC station metadata (steps 3A.1, 3B.2)
apiVersion: v3.0.2
kind: DataContract
id: project3-source-ndbc-stations
name: NOAA NDBC station metadata
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: The stations (id, position, other metadata) that make up the region's buoy network; needed to choose the
region's stations (3A.1), find the nearest buoy to a point and keep station metadata as SCD Type 2 rows (3B.2).
  limitations: >
    The build plan does not name the NDBC product that lists stations; it is identified during 3A.1 (design doc OQ-7).
    Field names beyond id and position are not confirmed.
  usage: Read by C1 (station selection) and C12 (SCD Type 2 merge).
team:
  - username: john
    role: owner
support:
  - channel: github-issues
```

```
    url: "issue: to be created"
servers:
  - server: ndbc-stations
    type: custom
    description: To be identified (OQ-7).
schema:
  - name: ndbc_station_listing
    physicalName: station list (to be identified, OQ-7)
    properties:
      - name: station_id
        logicalType: string
        required: true
        primaryKey: true
      - name: latitude
        logicalType: number
        required: true
      - name: longitude
        logicalType: number
        required: true
      - name: station_attributes
        logicalType: object
        required: false
        description: Further metadata fields (for example name, owner and type); the exact fields are confirmed at
first load. A change in these creates a new SCD Type 2 row downstream.
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Read by the nightly run; change detection produces SCD Type 2 rows.
customProperties:
  - property: schemaDiffBeforeLoad
    value: true
```

Ocean Map/contracts/table-ais-messages.yaml

```
# ODCS data contract: lake table. ais_messages (Iceberg) (step 3B.1)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-ais-messages
name: ais_messages
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Parsed AISStream messages written by the streaming job (live and replay), one row per event.
  limitations: >
    Only fields confirmed in the AISStream format are listed: MessageType, MetaData (MMSI, ship name, position) and the
typed Message.
    Typed fields (IMO number and other static data) will be added when the message types used are confirmed (OQ-13).
    The source-key composition and watermark are open (OQ-14). Public display of any vessel field is restricted (ADR
0008).
  usage: Input to vessel identity resolution (C14); counts for the lookup. Never exposed directly to the public.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    description: Iceberg table in the Glue database for the environment.
schema:
  - name: ais_messages
    physicalType: iceberg-table
    properties:
      - name: source_key
        logicalType: string
        required: true
        primaryKey: true
        description: Deduplication key (composition open, OQ-14).
      - name: message_type
        logicalType: string
        required: true
        description: AISStream MessageType.
```

```
      - name: mmsi
        logicalType: string
        required: false
        description: From MetaData.
      - name: ship_name
        logicalType: string
        required: false
        description: From MetaData.
      - name: latitude
        logicalType: number
        required: false
        description: From MetaData position.
      - name: longitude
        logicalType: number
        required: false
        description: From MetaData position.
      - name: event_time_utc
        logicalType: timestamp
        required: true
        description: Event time (platform rule 2). Source field to be confirmed (OQ-13).
      - name: message_body
        logicalType: object
        required: false
        description: The typed Message kept as semi-structured data until typed columns are defined (OQ-13).
      - name: origin
        logicalType: string
        required: true
        description: live or capture_replay.
      - name: capture_part_uri
        logicalType: string
        required: false
        description: Raw capture file that held the message (traceability to raw).
      - name: ingested_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: library
        metric: duplicateValues
        arguments:
          properties: [source_key]
        mustBe: 0
        description: Replaying a capture twice writes each event once (3B.1 Done when).
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: The nightly replay processes the day's capture; live sessions add rows on demand.
  - property: freshness
    value: "open"
    description: OQ-8.
customProperties:
  - property: streamingPattern
    value: "explicit schema, S3 checkpoints, watermark, dropDuplicatesWithinWatermark, availableNow for replay (ADR
0007)"
```

Ocean Map/contracts/table-buoy-flags.yaml

```
# ODCS data contract: lake table. buoy_flags (Iceberg) (step 3B.6)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-buoy-flags
name: buoy_flags
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Flags for stuck, spiking or silent buoys with the rule that fired.
  limitations: Thresholds and model are open (OQ-20). There is no generated text (ADR 0014).
  usage: Written by C18; read by the lookup (nearest buoy flags) and the data-quality agent.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
```

```
servers:
  - server: lake
    type: custom
    description: Iceberg table in the Glue database for the environment.
schema:
  - name: buoy_flags
    physicalType: iceberg-table
    properties:
      - name: station_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: flagged_at_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: flag_type
        logicalType: string
        required: true
        description: stuck, spiking or silent.
      - name: rule_id
        logicalType: string
        required: true
        description: The rule that fired.
      - name: model_score
        logicalType: number
        required: false
        description: scikit-learn model output when used as a second signal (OQ-20).
      - name: reading_window_start_utc
        logicalType: timestamp
        required: true
      - name: reading_window_end_utc
        logicalType: timestamp
        required: true
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: library
        metric: nullValues
        arguments:
          properties: [station_id, flagged_at_utc, flag_type, rule_id]
        mustBe: 0
slaProperties:
  - property: frequency
    value: 1
    unit: d
```

Ocean Map/contracts/table-buoy-readings.yaml

```
# ODCS data contract: published/lake table. buoy_readings (Iceberg)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-buoy-readings
name: buoy_readings
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Conformed buoy readings, realtime (3A.1) and history (3B.2), one row per station and observation time.
  limitations: Column names and units are confirmed at first load (OQ-7). Faulty buoys are flagged in buoy_flags, not
deleted from this table.
  usage: Input to dbt staging (stg_buoy_readings), fault detection (C18) and the latest-conditions table.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    environment: dev-and-live-separate
```

```
    description: Iceberg table in the Glue database for the environment. Names follow platform conventions (OQ-27).
schema:
  - name: buoy_readings
    physicalType: iceberg-table
    properties:
      - name: station_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: observation_time_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Event time in UTC (platform rules 2, 15).
      - name: wave_height
        logicalType: number
        required: false
      - name: water_temperature
        logicalType: number
        required: false
      - name: source_key
        logicalType: string
        required: true
        description: Source key identifying the record in the raw file (platform rule 2).
      - name: raw_file_uri
        logicalType: string
        required: true
        description: Link to the immutable raw file (with hash and fetch time).
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
        description: Load time.
      - name: origin
        logicalType: string
        required: true
        description: realtime or history.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
        arguments:
          properties: [station_id, observation_time_utc, source_key, loaded_at_utc]
      - type: custom
        engine: great-expectations
        description: Range checks on wave_height and water_temperature (bounds open, OQ-7).
slaProperties:
  - property: frequency
    value: 1
    unit: d
  - property: freshness
    value: "open"
    description: Numeric target open (OQ-8).
customProperties:
  - property: partitioning
    value: "overwrite by data interval (realtime) or station-year (history); never blind-append"
  - property: icebergMaintenance
    value: "compact, expire snapshots, remove orphan files as last nightly task"
```

Ocean Map/contracts/table-buoy-stations.yaml

```
# ODCS data contract: lake table. buoy_stations (Iceberg, SCD Type 2) (step 3B.2)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-buoy-stations
name: buoy_stations
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Station metadata with history, one row per station per period during which its metadata was unchanged.
  limitations: Attribute fields are confirmed at first load (OQ-7).
  usage: Nearest-buoy calculation, fault detection, Snowflake marts.
team:
```

```
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    description: Iceberg table in the Glue database for the environment.
schema:
  - name: buoy_stations
    physicalType: iceberg-table
    properties:
      - name: station_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: valid_from_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: valid_to_utc
        logicalType: timestamp
        required: false
        description: Null for the current row.
      - name: is_current
        logicalType: boolean
        required: true
      - name: latitude
        logicalType: number
        required: true
      - name: longitude
        logicalType: number
        required: true
      - name: h3_cell
        logicalType: string
        required: true
        description: H3 cell of the station at the resolution used for the nearest-buoy join (resolution set with OQ-6
and ADR 0002).
      - name: station_attributes
        logicalType: object
        required: false
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: custom
        engine: dbt
        description: Exactly one current row per station; no overlapping validity ranges.
slaProperties:
  - property: frequency
    value: 1
    unit: d
customProperties:
  - property: historyModel
    value: "SCD Type 2 (platform rule 8)"
```

Ocean Map/contracts/table-fishing-cells.yaml

```
# ODCS data contract: lake table. fishing_cells (Iceberg) (step 3A.3)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-fishing-cells
name: fishing_cells
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Per-cell counts of AIS fishing presence and of SAR detections with no AIS match ("possible dark vessels").
  limitations: >
    Counts per grid cell, not per vessel. Non-commercial data; credit required (ADR 0009). Unmatched SAR detections are
possible dark vessels, not confirmed (ADR 0004).
    Grid resolution and the H3 resolution used here are open (OQ-6).
```

```
  usage: Input to dbt (stg_fishing_cells) and the lookup.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    description: Iceberg table in the Glue database for the environment.
schema:
  - name: fishing_cells
    physicalType: iceberg-table
    properties:
      - name: h3_cell
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: H3 cell at the resolution chosen from the GFW grid (OQ-6).
      - name: report_date
        logicalType: date
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: layer
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 3
        description: ais_fishing or sar_unmatched.
      - name: fishing_hours
        logicalType: number
        required: false
        description: For layer ais_fishing.
      - name: detections
        logicalType: number
        required: false
        description: For layer sar_unmatched (possible dark vessels).
      - name: source_key
        logicalType: string
        required: true
      - name: pull_id
        logicalType: string
        required: true
        description: Identifies the pull; one partition per pull keeps history in rows.
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: library
        metric: nullValues
        arguments:
          properties: [h3_cell, report_date, layer, loaded_at_utc]
        mustBe: 0
      - type: custom
        engine: great-expectations
        description: fishing_hours and detections are non-negative; result is non-empty for the interval.
slaProperties:
  - property: frequency
    value: 1
    unit: d
  - property: freshness
    value: "open"
    description: Depends on the provider's data delay (OQ-6, OQ-8).
customProperties:
  - property: partitioning
    value: "overwrite by data interval; never blind-append"
```

Ocean Map/contracts/table-ocean-buoy-latest.yaml

```
# ODCS data contract: published table. ocean_buoy_latest (Parquet, gold) (steps 3A.4, 3B.6)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-ocean-buoy-latest
```

```yaml
name: ocean_buoy_latest
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Latest conditions per buoy station, plus fault flags with their rule and readings (Phase B), for the
nearest-buoy part of the lookup.
  limitations: >
    "Latest" means the latest reading loaded by the nightly run, so it can be up to about a day old unless the pull
frequency changes (OQ-7). When the run is late the page shows this data with a late notice and the last good run time
(ADR 0016; lateness rule OQ-29).
    Flags carry no generated text (ADR 0014).
  usage: Read by the lookup Lambda (DuckDB over Parquet) with the session box off.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: published-parquet
    type: s3
    environment: dev-and-live-separate
    description: Parquet files in the site/lake bucket for the environment.
schema:
  - name: ocean_buoy_latest
    physicalType: parquet
    properties:
      - name: station_id
        logicalType: string
        required: true
        primaryKey: true
      - name: latitude
        logicalType: number
        required: true
      - name: longitude
        logicalType: number
        required: true
      - name: observation_time_utc
        logicalType: timestamp
        required: true
        description: Time of the latest reading; shown on the page in local time (platform rule 15).
      - name: wave_height
        logicalType: number
        required: false
      - name: water_temperature
        logicalType: number
        required: false
      - name: flag_type
        logicalType: string
        required: false
        description: stuck, spiking or silent when a flag is active (Phase B).
      - name: flag_rule_id
        logicalType: string
        required: false
      - name: flag_readings
        logicalType: array
        required: false
        description: The readings shown with the flag.
      - name: published_at_utc
        logicalType: timestamp
        required: true
        description: Time of the last good publish; shown as the last good run time in the late notice (ADR 0016).
    quality:
      - type: library
        metric: duplicateValues
        arguments:
          properties: [station_id]
        mustBe: 0
      - type: custom
        engine: great-expectations
        description: Range checks on wave_height and water_temperature (bounds open, OQ-7); freshness check.
slaProperties:
  - property: frequency
    value: 1
```

```
    unit: d
  - property: freshness
    value: "open"
    description: OQ-8.
customProperties:
  - property: publishPattern
    value: "write, audit, publish (platform rule 4)"
```

Ocean Map/contracts/table-ocean-cell-summary.yaml

```
# ODCS data contract: published table. ocean_cell_summary (Parquet, gold) (step 3A.4)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-ocean-cell-summary
name: ocean_cell_summary
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Per-cell published answer inputs for the lookup (mapped flag, fishing activity, possible dark vessel
detections and the nearest buoy station).
  limitations: >
    Published only after the Great Expectations gate passes. Fishing counts come from non-commercial data with credit
(ADR 0009).
    "possible dark vessels" wording is fixed (ADR 0004). Counts only; no vessel names or MMSI (ADR 0008). The
neighborhood used for "nearby" is open (OQ-5).
  usage: Read by the lookup Lambda (DuckDB over Parquet) with the session box off.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: published-parquet
    type: s3
    environment: dev-and-live-separate
    description: Parquet files in the site/lake bucket for the environment.
schema:
  - name: ocean_cell_summary
    physicalType: parquet
    properties:
      - name: h3_cell
        logicalType: string
        required: true
        primaryKey: true
        description: H3 cell (resolution 8 for the seafloor part).
      - name: is_mapped
        logicalType: boolean
        required: true
      - name: mapped_pixel_share
        logicalType: number
        required: false
      - name: ais_fishing_hours
        logicalType: number
        required: false
        description: Aggregated to the cell via cell_to_parent (resolution per OQ-6).
      - name: possible_dark_vessel_detections
        logicalType: number
        required: false
        description: Unmatched SAR detections. Label "possible dark vessels".
      - name: nearest_station_id
        logicalType: string
        required: false
      - name: nearest_station_distance
        logicalType: number
        required: false
        description: Distance unit set at build.
      - name: fishing_data_as_of_utc
        logicalType: timestamp
        required: true
        description: Latest date included in the fishing counts.
      - name: published_at_utc
        logicalType: timestamp
        required: true
```

```
        description: Time of the last good publish; shown as the last good run time in the late notice (ADR 0016).
    quality:
      - type: library
        metric: duplicateValues
        arguments:
          properties: [h3_cell]
        mustBe: 0
      - type: custom
        engine: great-expectations
        description: Freshness, non-empty and value-range checks at the publish gate; counts are non-negative.
slaProperties:
  - property: frequency
    value: 1
    unit: d
  - property: freshness
    value: "open"
    description: OQ-8.
customProperties:
  - property: publishPattern
    value: "write, audit, publish (platform rule 4)"
```

Ocean Map/contracts/table-seafloor-cells.yaml

```
# ODCS data contract: lake table. seafloor_cells (Iceberg) (step 3A.2)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-seafloor-cells
name: seafloor_cells
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: One row per ocean cell in the region at H3 resolution 8, flagged mapped or unmapped from GEBCO's source-type
grid.
  limitations: >
    Mapped means directly measured (GEBCO codes 10 to 17, ADR 0003). Interpolated or predicted (40 to 46) and unknown
(70 to 72) are not mapped.
    Land (code 0) cells are excluded. How pixel codes combine into a cell flag is open (OQ-4).
  usage: Input to dbt (stg_seafloor_cells), unmapped-share calculation for the lookup.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    description: Iceberg table in the Glue database for the environment.
schema:
  - name: seafloor_cells
    physicalType: iceberg-table
    properties:
      - name: h3_cell
        logicalType: string
        required: true
        primaryKey: true
        description: H3 resolution 8 cell id.
      - name: is_mapped
        logicalType: boolean
        required: true
        description: True when the cell counts as mapped by the rule in ADR 0003 (the aggregation from pixels is open,
OQ-4).
      - name: mapped_pixel_share
        logicalType: number
        required: false
        description: Share of the cell's pixels with codes 10 to 17. Included so the unmapped share can be computed;
whether it is kept depends on OQ-4.
      - name: source_type_code_summary
        logicalType: object
        required: false
        description: Count of pixels per source-type code in the cell (supports audit of the rule).
      - name: gebco_release
        logicalType: string
        required: true
```

```
        description: Release identifier of the grid used (GEBCO 2026).
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: library
        metric: duplicateValues
        arguments:
          properties: [h3_cell]
        mustBe: 0
      - type: library
        metric: nullValues
        arguments:
          properties: [h3_cell, is_mapped]
        mustBe: 0
        description: Every ocean cell in the region has a mapped flag (3A.2 Done when).
slaProperties:
  - property: frequency
    value: "on release"
    description: Rebuilt when the GEBCO release or the region changes, not nightly.
customProperties:
  - property: h3Resolution
    value: 8
```

Ocean Map/contracts/table-vessel-golden.yaml

```
# ODCS data contract: lake table. vessel_golden and vessel_identity_link (Iceberg, SCD Type 2) (step 3B.3)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-vessel-golden
name: vessel_golden
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: The cleaned vessel identity table, with one current row per real vessel and history kept in rows.
  limitations: >
    Internal table. Vessel names and MMSI are never shown publicly (ADR 0008, decided by John, Oct 9, 2026). The rule
is loosened only after AISStream answers John's question about their terms (OQ-3). IMO and other static fields depend
on the AISStream
    message types used (OQ-13). Survivorship rules for choosing field values are set at build.
  usage: Output of C14; read by the lookup counts, the review page and Snowflake marts.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    description: Iceberg tables in the Glue database for the environment.
schema:
  - name: vessel_golden
    physicalType: iceberg-table
    properties:
      - name: vessel_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Stable key of the real vessel.
      - name: valid_from_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: valid_to_utc
        logicalType: timestamp
        required: false
      - name: is_current
        logicalType: boolean
        required: true
      - name: imo
        logicalType: string
```

```
        required: false
        description: IMO number when known and the check digit passes.
      - name: names
        logicalType: array
        required: false
        description: Normalized and original name variants.
      - name: match_confidence
        logicalType: number
        required: false
      - name: matcher_version
        logicalType: string
        required: true
        description: rules, rules_plus_ai or classifier (ADR 0012).
      - name: evidence
        logicalType: object
        required: false
        description: Rules that fired and label ids that support the record.
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: custom
        engine: dbt
        description: Exactly one current row per vessel_key; no overlapping validity ranges.
  - name: vessel_identity_link
    physicalType: iceberg-table
    description: Maps each MMSI (and name variant) to a vessel_key, with validity.
    properties:
      - name: mmsi
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: vessel_key
        logicalType: string
        required: true
      - name: valid_from_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: valid_to_utc
        logicalType: timestamp
        required: false
      - name: mmsi_status
        logicalType: string
        required: true
        description: valid, invalid_structure, unknown_country_prefix or collision (one MMSI used by two ships at
once).
slaProperties:
  - property: frequency
    value: 1
    unit: d
customProperties:
  - property: historyModel
    value: "SCD Type 2; labels applied as new rows (platform rule 8)"
```

Ocean Map/contracts/table-vessel-labels.yaml

```
# ODCS data contract: lake table. vessel_labels (Iceberg) (steps 3B.3, 3B.5)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-vessel-labels
name: vessel_labels
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: John's decisions on review items and agent proposals, copied nightly from DynamoDB; applied as SCD Type 2
rows and used as training labels.
  limitations: Contains no visitor data. John, the owner, is the only person who records decisions.
  usage: Read by C14 (apply_labels_scd2), C16 (training set) and the proposal log.
team:
  - username: john
    role: owner
```

```
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    description: Iceberg table in the Glue database for the environment.
schema:
  - name: vessel_labels
    physicalType: iceberg-table
    properties:
      - name: label_id
        logicalType: string
        required: true
        primaryKey: true
      - name: item_id
        logicalType: string
        required: true
        description: Review queue item.
      - name: item_kind
        logicalType: string
        required: true
        description: match_review or agent_proposal.
      - name: pair_id
        logicalType: string
        required: false
      - name: decision
        logicalType: string
        required: true
        description: Approved, rejected or the decision values defined by the review page.
      - name: decided_at_utc
        logicalType: timestamp
        required: true
      - name: copied_at_utc
        logicalType: timestamp
        required: true
        description: Time of the nightly copy from DynamoDB.
slaProperties:
  - property: frequency
    value: 1
    unit: d
```

Ocean Map/contracts/table-vessel-pairs.yaml

```
# ODCS data contract: lake table. vessel_pairs (Iceberg) (steps 3B.3, 3B.4)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-vessel-pairs
name: vessel_pairs
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Candidate vessel pairs with each matcher version's score, route and (for version 2) the LLM decision with
reason and confidence; the basis of the three-version comparison.
  limitations: >
    Includes the 200 hand-labeled evaluation pairs, which must be flagged and kept out of training (OQ-17).
    Thresholds and the embedding model are open (OQ-16).
  usage: Written by C14 and C16; read by evaluation and comparison, and by classifier training.
team:
  - username: john
    role: owner
support:
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: lake
    type: custom
    description: Iceberg table in the Glue database for the environment.
schema:
  - name: vessel_pairs
    physicalType: iceberg-table
    properties:
      - name: pair_id
        logicalType: string
```

```
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: matcher_version
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: rules, rules_plus_ai or classifier.
      - name: left_id
        logicalType: string
        required: true
      - name: right_id
        logicalType: string
        required: true
      - name: candidate_source
        logicalType: string
        required: true
        description: rules or embedding.
      - name: score
        logicalType: number
        required: false
      - name: route
        logicalType: string
        required: true
        description: auto_match, review or no_match.
      - name: llm_match
        logicalType: boolean
        required: false
        description: Version 2 only. LLM decision.
      - name: llm_reason
        logicalType: string
        required: false
      - name: llm_confidence
        logicalType: number
        required: false
      - name: prompt_version
        logicalType: string
        required: false
        description: Prompt version in git.
      - name: is_evaluation_pair
        logicalType: boolean
        required: true
        description: True for the 200 hand-labeled pairs; excluded from training.
      - name: created_at_utc
        logicalType: timestamp
        required: true
slaProperties:
  - property: frequency
    value: 1
    unit: d
```

Ocean Map/contracts/table-vessel-review-queue.yaml

```
# ODCS data contract: DynamoDB table. vessel_review_queue (steps 3B.3, 3B.5)
apiVersion: v3.0.2
kind: DataContract
id: project3-table-vessel-review-queue
name: vessel_review_queue
version: 0.1.0
status: draft
domain: ocean-map
dataProduct: project3-ocean-map
description:
  purpose: Open and decided review items (low-confidence vessel matches and data-quality agent proposals).
  limitations: >
    Always-on store, writable while the session box is off. Point-in-time recovery on. Provisioned within the free
tier, so heavy traffic is throttled rather than billed.
    Owner only, behind one API key held in Parameter Store, keyed REST type; the page that reads it is not linked from
the public site (ADR 0015, decided by John, Oct 9, 2026).
  usage: Written by C14 (enqueue_for_review) and C17 (create_proposal, the agent's only write); read and decided
through the review page (C15); copied nightly to vessel_labels.
team:
  - username: john
    role: owner
support:
```

```
  - channel: github-issues
    url: "issue: to be created"
servers:
  - server: dynamodb
    type: dynamodb
    environment: dev-and-live-separate
    description: One table per environment. Key design (partition and sort keys) is set at build.
schema:
  - name: vessel_review_queue
    physicalType: dynamodb-table
    properties:
      - name: item_id
        logicalType: string
        required: true
        primaryKey: true
      - name: kind
        logicalType: string
        required: true
        description: match_review, merge_vessels, mark_buoy_faulty or open_issue.
      - name: status
        logicalType: string
        required: true
        description: open, approved or rejected.
      - name: payload
        logicalType: object
        required: true
        description: What is being reviewed (pair, vessels, buoy or issue text).
      - name: evidence
        logicalType: object
        required: true
        description: Rules that fired, scores, LLM reason and confidence, or the failed check.
      - name: created_by
        logicalType: string
        required: true
        description: pipeline or data_quality_agent.
      - name: created_at_utc
        logicalType: timestamp
        required: true
      - name: decision
        logicalType: string
        required: false
      - name: decided_at_utc
        logicalType: timestamp
        required: false
    quality:
      - type: custom
        engine: iam-test
        description: The agent role can create items with created_by data_quality_agent and cannot modify decision or
status fields.
slaProperties:
  - property: frequency
    value: "continuous"
    description: Writable at any hour.
customProperties:
  - property: pointInTimeRecovery
    value: true
  - property: nightlyCopy
    value: "to vessel_labels in the lake"
```

Part 9: API specification

Ocean Map/openapi.yaml

```
openapi: 3.0.3
info:
  title: Project 3 Ocean map API
  version: 0.1.0
  description: |
    Endpoints for the Ocean map project. Draft for John's review.
    Route names are provisional (design doc OQ-10). The OpenAPI version and the serving-template conventions
    come from the platform documents (../Platform/design-doc.md, OQ-27).

    Part 1: the public lookup (HTTP API type of API Gateway, FastAPI on Lambda, DuckDB over published Parquet).
    It collects no visitor data and answers with the session box off.

    Part 2: the review endpoints used by the review queue page. They write John's decisions. Decided by John, Oct 9,
2026:
    owner only, behind one API key held in Parameter Store, keyed REST type, page not linked from the public site (ADR
0015).
    API Gateway answers a missing or invalid key with 403 before the function runs; 401 in the responses below is kept
for
    a key the function itself rejects.

    The first request after a period of inactivity may take up to 10 seconds while the service initializes.
servers:
  - url: https://example.invalid/ocean
    description: Placeholder. The real base URL is set by the platform serving template. A dev and a live server exist.
tags:
  - name: lookup
    description: Public point lookup.
  - name: review
    description: Owner-only endpoints behind one API key (ADR 0015).
paths:
  /lookup:
    get:
      tags: [lookup]
      security: []
      operationId: lookupPoint
      summary: Answer for an ocean or coastal point
      description: |
        Converts the point to an H3 resolution 8 cell and returns how much nearby seafloor is unmapped,
        fishing activity nearby, and the nearest buoy's latest conditions, with credits and the times the data was
taken.
        "Nearby" is defined by the neighborhood rule decided in OQ-5. Land points and points outside the covered region
are rejected.
        Reads published Parquet only; never calls any upstream source.
      parameters:
        - name: lat
          in: query
          required: true
          description: Latitude in degrees (WGS 84).
          schema:
            type: number
            minimum: -90
            maximum: 90
        - name: lon
          in: query
          required: true
          description: Longitude in degrees (WGS 84).
          schema:
            type: number
            minimum: -180
            maximum: 180
      responses:
        '200':
          description: Lookup answer.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/LookupAnswer'
        '400':
          description: Missing or invalid lat or lon.
          content:
            application/json:
```

```
              schema:
                $ref: '#/components/schemas/Error'
        '404':
          description: The point is on land or outside the covered region (Phase A covers one region, ADR 0001).
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '429':
          description: Throttled by the gateway (limits open, OQ-10).
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '503':
          description: Published data unavailable or the function failed. The page shows an error state.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
  /review/items:
    get:
      tags: [review]
      security:
        - ReviewApiKey: []
      operationId: listReviewItems
      summary: List review items
      description: Lists review queue items (vessel match reviews and data-quality agent proposals), open first. Owner
only.
      parameters:
        - name: status
          in: query
          required: false
          schema:
            type: string
            enum: [open, approved, rejected]
        - name: cursor
          in: query
          required: false
          description: Opaque cursor from a previous response.
          schema:
            type: string
        - name: limit
          in: query
          required: false
          description: Page size. Maximum is set at build.
          schema:
            type: integer
            minimum: 1
      responses:
        '200':
          description: A page of review items.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ReviewItemPage'
        '401':
          description: Missing or invalid API key.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '403':
          description: Key not accepted (owner only).
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '429':
          description: Throttled.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
  /review/items/{item_id}:
    get:
```

```
    tags: [review]
    security:
      - ReviewApiKey: []
    operationId: getReviewItem
    summary: One review item with its evidence
    parameters:
      - $ref: '#/components/parameters/ItemId'
    responses:
      '200':
        description: The item.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ReviewItem'
      '401':
        description: Missing or invalid API key.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
      '403':
        description: Key not accepted (owner only).
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
      '404':
        description: No such item.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
  /review/items/{item_id}/decision:
    post:
      tags: [review]
      security:
        - ReviewApiKey: []
      operationId: decideReviewItem
      summary: Record the owner's decision
      description: |
        Stores the decision on the item. This is the only write the API performs. It does not itself change any vessel
        or buoy table. The nightly run copies decisions to the lake as labels and applies them as SCD Type 2 rows.
        Deciding an already decided item returns 409.
      parameters:
        - $ref: '#/components/parameters/ItemId'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/DecisionRequest'
      responses:
        '200':
          description: Decision stored.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ReviewItem'
        '400':
          description: Invalid decision value.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '401':
          description: Missing or invalid API key.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '403':
          description: Key not accepted (owner only).
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
```

```
        '404':
          description: No such item.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '409':
          description: The item already has a decision.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '429':
          description: Throttled.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
components:
  securitySchemes:
    ReviewApiKey:
      type: apiKey
      in: header
      name: x-api-key
      description: The one API key (owner only), held in Parameter Store. Keyed REST type of API Gateway.
  parameters:
    ItemId:
      name: item_id
      in: path
      required: true
      schema:
        type: string
  schemas:
    LookupAnswer:
      type: object
      required: [point, cell, seafloor, fishing, buoy, credits, data_times, late]
      properties:
        point:
          type: object
          required: [lat, lon]
          properties:
            lat:
              type: number
            lon:
              type: number
        cell:
          type: string
          description: H3 resolution 8 cell of the point.
        seafloor:
          type: object
          required: [unmapped_share, mapped_rule]
          properties:
            unmapped_share:
              type: number
              minimum: 0
              maximum: 1
              description: Share of nearby seafloor that is not directly measured (derivation open, OQ-4, OQ-5).
            mapped_rule:
              type: string
              description: Text stating that codes 10 to 17 of GEBCO's source-type grid count as mapped (ADR 0003).
        fishing:
          type: object
          required: [ais_fishing_hours, possible_dark_vessel_detections, label]
          properties:
            ais_fishing_hours:
              type: number
              nullable: true
            possible_dark_vessel_detections:
              type: number
              nullable: true
            label:
              type: string
              description: The fixed label wording, "possible dark vessels" (ADR 0004).
        buoy:
          type: object
          nullable: true
```

```
          description: Nearest buoy's latest conditions. Null when no buoy applies.
          properties:
            station_id:
              type: string
            distance:
              type: number
              description: Distance to the station. Unit fixed at build.
            observation_time_utc:
              type: string
              format: date-time
            wave_height:
              type: number
              nullable: true
            water_temperature:
              type: number
              nullable: true
            flag:
              $ref: '#/components/schemas/BuoyFlag'
        late:
          type: object
          required: [is_late, last_good_run_utc]
          description: Late-data display rule (ADR 0016). Old data is always returned; when late, the page shows a
visible notice with the last good run time.
          properties:
            is_late:
              type: boolean
              description: How lateness is determined is open (OQ-29).
            last_good_run_utc:
              type: string
              format: date-time
            notice:
              type: string
              nullable: true
              description: Notice text when late (wording open, OQ-29).
        credits:
          type: array
          items:
            type: string
          description: Credit lines for every source used (wording open, OQ-24).
        data_times:
          type: object
          description: When each part of the data was taken, in UTC. The page converts to local time.
          properties:
            fishing_as_of_utc:
              type: string
              format: date-time
            buoy_as_of_utc:
              type: string
              format: date-time
            published_at_utc:
              type: string
              format: date-time
    BuoyFlag:
      type: object
      nullable: true
      description: Present only in Phase B when a flag is active. Contains no generated text (ADR 0014).
      required: [flag_type, rule_id, readings]
      properties:
        flag_type:
          type: string
          enum: [stuck, spiking, silent]
        rule_id:
          type: string
        readings:
          type: array
          items:
            type: object
            properties:
              observation_time_utc:
                type: string
                format: date-time
              wave_height:
                type: number
                nullable: true
              water_temperature:
                type: number
```

```
              nullable: true
    ReviewItemPage:
      type: object
      required: [items]
      properties:
        items:
          type: array
          items:
            $ref: '#/components/schemas/ReviewItem'
        next_cursor:
          type: string
          nullable: true
    ReviewItem:
      type: object
      required: [item_id, kind, status, payload, evidence, created_by, created_at_utc]
      properties:
        item_id:
          type: string
        kind:
          type: string
          enum: [match_review, merge_vessels, mark_buoy_faulty, open_issue]
        status:
          type: string
          enum: [open, approved, rejected]
        payload:
          type: object
          additionalProperties: true
        evidence:
          type: object
          additionalProperties: true
        created_by:
          type: string
          enum: [pipeline, data_quality_agent]
        created_at_utc:
          type: string
          format: date-time
        decision:
          type: string
          nullable: true
        decided_at_utc:
          type: string
          format: date-time
          nullable: true
    DecisionRequest:
      type: object
      required: [decision]
      properties:
        decision:
          type: string
          enum: [approve, reject]
          description: Decision values for match reviews may need a third value such as "different vessel"; to be
settled with the review page design (OQ-19).
        note:
          type: string
          maxLength: 1000
          description: Optional note. The length limit is a placeholder to be confirmed at build.
    Error:
      type: object
      required: [error]
      properties:
        error:
          type: object
          required: [code, message]
          properties:
            code:
              type: string
            message:
              type: string
```

Part 10: Changelog

Ocean Map/changelog.md

Changelog: Project 3, Ocean map

Dates are the day of the change. Counts are checked against the open-questions table in design-doc.md section 16.

Oct 9, 2026: defaults confirmed

- John confirmed ADRs 0020 to 0023 (held-out evaluation pairs, latest buoy reading wording, IP addresses removed from logs, licence terms checked on provider pages). Their status is now Accepted.

Oct 9, 2026: revision 3 (project documents completed; platform links made real)

Documents added to the project folder

- README.md: what the project is, status, the document list, accounts and keys needed, credit and licences, and where open questions live.
- operations.md: schedules, freshness, monitoring, cost limits (including the Global Fishing Watch request limits, verified on the provider's page), runbooks to expect, backups and routine checks. Links to the platform's runbooks.
- security-and-privacy.md: what the project holds and how long it is kept, keys and roles, threats, and the licence and credit register with a status for each source.
- model-card.md: the parts that use models, intended and unintended use, evaluation, known limits, and safety and privacy.

New ADRs

- ADR 0022: IP addresses are removed from request logs before they are kept (default; the same rule as Project 1).
- ADR 0023: licence and credit wording is checked on the provider's own page; unverified entries are marked (default).

Links and answers

- Every reference to the platform now links to the platform set. The platform was "not yet written" in revision 2; it is now a draft awaiting approval. The platform set is being rewritten at the same time, so its ADR numbers are not cited here.
- The design doc said the `llm_calls` table is defined in the platform contracts. It is not there yet; the design doc now says so.
- OQ-24 is partly answered: Global Fishing Watch and GEBCO terms were read on their own pages. AISStream, NDBC, MarineCadastre's licence text and GEBCO's credit wording are unverified.
- OQ-27 is partly answered: the platform uses ODCS v3.0.2. The naming convention and contract template are still open.

Verified against primary sources (Oct 9, 2026)

- GEBCO terms of use (gebco.net): the grid is in the public domain and free to use; the source must be acknowledged; no endorsement may be implied. The exact credit wording is in the documentation with the grid, and was not read.
- Global Fishing Watch license page: CC BY-NC 4.0, non-commercial only; "Powered by Global Fishing Watch." is one accepted credit; 50,000 requests a day and 1,500,000 a month per user.
- AISStream: the provider's site returned HTTP 403; its terms are unverified.
- NDBC: no data policy or credit statement found on the NDBC home page; the public-domain statements found elsewhere are secondary only.
- MarineCadastre: the hub page gave no licence text; the CC0 statement is from a NOAA repository (secondary only).

Counts (section 16)

- 30 questions: 2 answered (OQ-11, OQ-26), 7 partly answered (OQ-3, OQ-17, OQ-18, OQ-21, OQ-24, OQ-27, OQ-29), 21 open.
- ADRs: 23 (ADR 0001 to ADR 0023). Proposed: 0001, 0007, 0012, 0018. Default: 0020, 0021, 0022, 0023. Accepted: the other 15.

Still for John

- Approve the four document types (requirements, design doc, ADRs, interface specs) and the new documents above.
- Decide whether Phase A may publish NDBC data before NDBC's own policy page is read (security-and-privacy.md, section 6).
- Ask AISStream for its terms (OQ-3, OQ-24).
- Check whether the AISStream-like key published on a third-party page is his; if it is, rotate it.

Oct 9, 2026: revision 2 (answered decisions applied; checks and fixes)

Answered decisions written into the documents

- Late or failed nightly run: old data with a late notice and the last good run time (ADR 0016). Already in the set; cross-references checked. The late threshold and notice wording are open (OQ-29).
- Review queue: owner only, behind one API key, page not linked from the public site (ADR 0015). Answers OQ-11. How the key reaches the page is still open (OQ-31).
- AISStream: maps and counts only; vessel names and MMSI internal (ADR 0008). The terms reply from AISStream is still open (OQ-3, OQ-24).
- Classifier training: manual export from Databricks Free Edition (ADR 0012). Whether the export can be automated is still open (OQ-18).
- Metrics, evals, tracing and MCP: proposed step list written (requirements.md section 8; ADR 0018). Approval is pending (OQ-21, OQ-30).
- Phase C done-when checks: recorded as missing until Phase C is designed (ADR 0019). Answers OQ-26.
- MMSI country prefix reference: stays open; no list is guessed (ADR 0017; OQ-15).

New ADRs

- ADR 0017: MMSI prefix reference stays open. Accepted (John's decision).
- ADR 0018: metrics, evals, tracing and MCP step list. Proposed, awaiting John's approval.
- ADR 0019: Phase C done-when checks recorded as missing. Accepted (John's choice).
- ADR 0020: the 200 evaluation pairs are never used for training or prompts (FR-57). Default; John's approval requested (OQ-30).
- ADR 0021: say "latest buoy reading" with its time, not "current conditions". Default.

Corrections found in checking

- Section 16 kept its answered questions in the table, with an answer column and ADR number, as the documentation rules require. Rows OQ-11 and OQ-26 were cited elsewhere but missing from the table; both are restored. The section 16 intro now says answered rows stay.
- The buoy risk row said "current conditions"; it now says "latest reading" (ADR 0021).
- FR-32, FR-57, the step list and the Phase C note now cite their ADRs.
- Section 3.4 of the design doc lists every answered decision with its ADR number.
- This folder had no changelog; this file was created.

Verified against primary sources (Oct 9, 2026)

- No licence or terms wording was changed in this revision. The terms of AISStream, NDBC, GEBCO 2026 and Global Fishing Watch were not re-read in this pass and remain unverified (OQ-24).

Counts (section 16)

- Open questions: 30 rows. 2 answered (OQ-11, OQ-26), 5 partly answered (OQ-3, OQ-17, OQ-18, OQ-21, OQ-29), 23 open.
- ADRs: 21 (ADR 0001 to ADR 0021). Proposed: 0001, 0007, 0012, 0018. Default: 0020, 0021. Accepted: the other 15.

Still for John

- Approve or change the step list (ADR 0018), the defaults in ADR 0020 and 0021, and FR-20 (OQ-30).
- Decide how the review key reaches the page (OQ-31) before the review page is built.
- Ask AISStream about its terms (OQ-3, OQ-24).
- Confirm one question that the decisions list cites but this set does not define. The list says it is about missing MLflow steps, but the project 3 documents never mention MLflow. The number was left out of this table until John confirms it.
- Confirm the placeholder links to `../Platform/`. The platform documents are not written, so those links do not resolve yet.
- The set has no README.md (setup step 1 lists one). It was not created in this revision.

Before revision 2

- No changelog was kept for this set. The earlier PDF, "Project 3 Ocean Map.pdf", was removed as an old version on Oct 9, 2026. The current PDF is `full-documentation.pdf` in this folder (revision 3).